Senior DevOps Engineer –EKS- Parra – Mackay Executive Pty Ltd – Hamiltom etc. -12/6/25
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Full Stack Engineer /Integration Solution Architect /Software Engineer/ Python/R Developer/ Salesforce/Mulesoft /AI/ML/BI/SharePoint/Dynamics 365 Developer /EDW/DBA –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps Engineer/Integration Architect/Full Stack (SF/MuleSoft/Py/AI/ML) Developer/Snr Data Engineer
Responsibilities:

- Highly skilled Senior DevOps Engineer passionate about driving development productivity, automation, quality, and continuous improvement, responsible for leading and developing internal DevOps capabilities while consistently employing best practices & standards-compliant techniques for the long-term operation, monitoring, & upgrade of enterprise software.
- Architected and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for secure data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power Automate,Power App, Power BI) & AI Builder for D365 into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Automated enterprise software delivery and quality assurance in production environments by designing and implementing CI/CD pipelines using AWS CodePipeline, CodeBuild, and CodeDeploy, alongside Bamboo CI or GitHub Actions.
- Proficient in Infrastructure-as-Code (IaC), leveraging the AWS Cloud Development Kit (CDK) to define cloud infrastructure in code, and Troposphere, a Python library for generating CloudFormation templates, to automate infrastructure provisioning.
- Possesses solid coding/scripting skills in Java, Python, and Bash, enabling dynamic solution development and automation.
- Crafted AI-driven CI/CD pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Utilized AWS SDK for Java to programmatically trigger a CodeBuild project. E.g: codeBuildClient.startBuild(new StartBuildRequest().withProjectName("MyCICDProject"));
- Used Bash for advanced DevOps automation, dynamically adjusting AWS security groups to allow ephemeral IP ranges (e.g., TEMP_IP=$(curl ... | jq -r '.ip_address')) for post-deployment validation with IP details fetched from external sources.
- Linux Server Provisioning: Leveraged Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated EC2 instance creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Jenkins or Bamboo CI
- Championed efficient & scalable deployment of containerized applications by utilizing AWS Serverless container orchestration platforms like Amazon EKS & ECS Fargate (for serverless compute), enabling streamlined operation of Docker workloads in production environments. This included orchestrating complex big data processing workflows with Apache Airflow, leveraging Cloud Native Distribution Infrastructure (CNDI) for robust & scalable ETL/ELT data pipelines.
- Expertise extends to extensive DevOps support on AWS, including backup/recovery, monitoring with CloudWatch, and alerting, alongside a solid understanding of security principles, DDoS mitigation techniques, IPS/IDS, and Web Application Firewalls to safeguard infrastructure. Participating in digital transformation and capability uplift.

- Possess solid cloud experience across AWS core services, including IAM for access management, VPC for network isolation, EC2 for scalable compute, RDS for managed databases, Route53 for DNS, CloudFront for content delivery, CloudWatch for monitoring, Lambda for serverless functions, and DynamoDB for NoSQL data storage.
- Leveraged AWS Elastic Beanstalk for simplified, automated deployments of web applications, effectively abstracting infrastructure management while ensuring high availability, automatic scaling, and streamlined updates to accelerate development cycles and reduce operational overhead.
- Managed comprehensive cloud architectures on AWS, delivering high-performance solutions that integrate React/Redux applications with Mulesoft-driven APIs, utilizing both SQL and NoSQL databases for persistent data storage.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., dfx force:source:pull) from Salesforce organizations.
- AI-focused DevSecOps Engineer specializing in securing complex AI platforms and APIs within AWS and Mulesoft Anypoint Platform environments, leveraging expertise in observability, managing custom API Gateways (e.g., OKAPIGW), implementing robust CI/CD pipelines, and ensuring stringent enterprise network security.
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline, and isolated local AI Agents, including Security risk control AI agents, leveraging Ollama 4 and Google NotebookLM. These agents incorporate the latest AI technology, Model Context Protocol (MCP), and utilize RAG (Retrieval Augmented Generation) via an Agent-to-Agent communication protocol for advanced data analysis and decision-making.
- Designed and implemented AI-driven workflows using n8n to automate client intake, medical history triage, and wellness reporting for aged care service, integrating AI agents & LLMs to enhance operational efficiency & personalized aged care.
- Developed AI APIs on the NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabeled data and supervised KNN within R for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr DevOps/Integration Solution Architect/Full Stack Developer(SF/MuleSoft/AI)/Snr Data Engineer/EDW
Responsibilities:

- Integration Solution Architect specialized in designing and implementing complex integrations between modern cloud-based software (Azure, AWS, GCP, Mulesoft) and legacy on-site applications using IPasS and Biztalk.
- Designed and deployed a resilient hybrid cloud architecture spanning AWS, GCP, and Azure, establishing secure IPsec VPN tunnels between VPCs/VNets to safeguard critical RBA financial data transfers and maintain compliance.
- Architected and implemented enterprise-grade AWS Landing Zones, leveraging IaC (CloudFormation/Terraform) and AWS Control Tower for stringent multi-account governance and CloudWatch for deep operational insights.
- Integrated diverse M365 environments (O365, MS Teams, SharePoint Online, Dynamics 365, MS Flow, Power App, Power BI) within these secure zones, delivering a cohesive and compliant platform optimized for robust Microsoft workload support.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments using MuleSoft Anypoint Studio 7 and Mule 4 .
- Architected and implemented robust CI/CD pipelines utilizing GitHub Actions for continuous integration and artifact generation, seamlessly deploying applications to AWS Elastic Beanstalk to leverage its simplified infrastructure management and efficient scaling capabilities.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- Integrated M365 environments (O365, MS Teams, SharePoint online, Dynamics 365, Power App, Microsoft Flow) into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless failover between AWS and Azure. This ensures resilience against outages.
- Architected & managed scalable cloud solutions on AWS, ensuring high-performance integration of React/Redux front-ends with critical Mulesoft-driven APIs. Expertise includes leveraging diverse SQL & NoSQL databases for secure, persistent data storage in complex financial contexts.
- Design, develop, and maintain automated ETL/ELT data pipeline/API orchestration using either:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Linux Server Provisioning: Leveraged Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated EC2 instance creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Jenkins or Bamboo CI.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …

- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Developed RBA financial trading-tailored LLMs (e.g., BERT, GPT-4, LLaMA, Falcon) & integrated ML/DL algorithms for financial data analysis with Python's quantitative libraries (e.g., NumPy, Pandas, Scikit-learn) & AWS SageMaker, enabling accurate & personalized financial insights & decision support.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Leveraged advanced Machine Learning techniques to identify & optimize peak values derived from the intersection of linear regression trends and complex sinusoidal patterns in multi-dimensional datasets, enabling enhanced quantitative analysis.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Ensured resilient and secure AWS operations through comprehensive DevOps support, implementing backup/recovery, proactive CloudWatch monitoring & alerting, and deploying advanced security controls such as DDoS mitigation, IPS/IDS, & Web Application Firewalls. Contributed significantly to enterprise digital transformation & capability uplift programs.
- Utilized AWS Glue Studio for data transformation, Apache Spark for data manipulation, AWS S3 for data repository, AWS Redshift for data storage, and MuleSoft Anypoint Platform for system connectivity, alongside proficiency in established technologies such as Teradata, Hadoop, Tableau, and SAS, and AWS-specific services like EMR and Redshift.
- Established and managed continuous integration/continuous delivery automation using GitLab CICD for RBA's Salesforce and MuleSoft platforms, facilitating accelerated release cycles. Deployed Git version control and connected heritage systems with AWS, Azure, and Salesforce cloud platforms through MuleSoft Anypoint APIs.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., dfx force:source:pull) from Salesforce organizations.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.Independently architecting a multi-region, fault-tolerant Kubernetes cluster to support critical applications.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr DevOps/Full Stack Developer(SF/MuleSoft/AI)/Snr Data Engineer /Salesforce Support Engineer/EDW
Responsibilities:

- Architected and implemented enterprise-grade AWS Landing Zones, leveraging IaC (CloudFormation/Terraform) and AWS Control Tower for stringent multi-account governance and CloudWatch for deep operational insights.
- Crucially, integrated diverse M365 environments (O365, MS Teams, SharePoint Online, Dynamics 365, MS Flow, Power App) within these secure zones, delivering a cohesive & compliant platform optimized for robust Microsoft workload support.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
- Managed comprehensive cloud architectures on AWS, delivering high-performance solutions that seamlessly integrate React/Redux applications with Mulesoft-driven APIs, utilizing both SQL & NoSQL databases for robust, persistent data storage within enterprise environments.
- Delivered extensive DevOps support on AWS, specializing in strengthening infrastructure security through backup/recovery, CloudWatch monitoring & alerting, and advanced defenses including DDoS mitigation, IPS/IDS, & Web Application Firewalls. Played a key role in driving digital transformation & capability uplift.
- Managed comprehensive cloud architectures on AWS, delivering high-performance solutions that seamlessly integrate React/Redux applications with Mulesoft-driven APIs, utilizing both SQL & NoSQL databases for robust, persistent data storage within enterprise environments.
- Built and implemented Medallion Architecture with Data Lake in AWS:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting. kinesis_client = boto3.client('kinesis', region_name='ap-southeast-2')
- Leveraged Matillion and ERwin to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs, within both enterprise data warehouse (Inmon) and data mart (Kimball) architectures.
- Engineered seamless data and workflow integrations leveraging MuleSoft Anypoint Platform, including API development and hybrid deployment strategies, and integrated with AWS API Gateway and Azure IPaaS.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
  - **Azure: Azure DevOps, Azure Pipelines**

- **AWS: AWS CodePipeline (CodeBuild/CodeDeploy), AWS Elastic Beanstalk**
- **GCP: Composer/Cloud Build/Cloud Deploy**
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Developed and optimized ETL/ELT data pipelines using AWS Glue, Python, SQL, and Apache Spark, migrating and transforming data between on-premises systems and cloud platforms, including Salesforce.
- Implemented secure artifact control and project oversight using Artifactory and Jira.
- Skilled in data modeling (SPARX), data warehousing (AWS Redshift), and data analytics (SageMaker, Python, R).
- Implemented and managed secure certificate lifecycle management (PKI) and integrated it with Salesforce.
- **Automated Certificate Lifecycle Management**: Developed and implemented automated certificate lifecycle management processes, including issuance, renewal, and revocation, to maintain a secure and compliant PKI infrastructure.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Full Stack Developer(DF/Mulesoft/Java/Py)/Data Migration/Software Test Engineer
Responsibilities:

- Engineered and implemented a robust multi-cloud hybrid architecture across AWS, GCP, and Azure, leveraging Cloud VPN Gateways and IPsec encryption to ensure secure, high-performance data transfer and seamless connectivity for financial operations.
- Developed and implemented ETL pipelines using AWS Glue, Python, and SQL to migrate Health Care Approved Providers' data into Salesforce, supporting a critical aged care sector transformation.
- Leveraged Amazon Redshift for high-performance data warehousing and analytics. Utilized Amazon EMR with PySpark to build data pipelines for extracting, transforming, and loading (ETL) data from diverse sources into a Data Lake. Implemented ETL processes using Apache Spark, integrating with AWS S3 for data storage and AWS Redshift for data warehousing.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- **Advanced Salesforce Support**: Delivered advanced Salesforce support, ensuring the seamless operation of critical university systems by proactively investigating and resolving complex technical challenges within OmniStudio components (FlexCards, OmniScripts, DataRaptors, Integration Procedures).
- **System Integrity and Administration**: Maintained robust system integrity through comprehensive administration, including user management, application metrics monitoring, configuration, and data integrity assurance."
- DevOps and Deployment Management: Implemented and managed deployments using a structured DevOps framework, contributing to efficient and reliable system updates.
- Troubleshooting and System Enhancement: Conducted thorough log analysis to diagnose and resolve system anomalies, enhancing system stability and performance. These efforts directly supported the university's ability to maintain efficient administrative & academic processes, ensuring reliable access to essential data & resources for faculty, staff, and students.
- Demonstrated ability to maintain and configure Salesforce Service Cloud and Health Cloud applications through declarative tools, ensuring data integrity and optimal system performance.
- Using either Python or Java, developed custom ETL/API orchestrations applicable to cloud and on-premise applications.
- Implemented automated testing frameworks (JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, AWS DevOps) to integrate development and testing processes. Utilized Git for version control, Maven and Gradle for building, & JUnit for automated testing.
- Implemented GitOps deployment strategy for efficient infrastructure and application management. Provisioned and managed AWS Kubernetes Service (EKS) clusters using Terraform and kubectl.
- PKI Integration and Testing: Designed and executed thorough testing of PKI integrations with cloud platforms and MuleSoft, validating certificate functionality and identifying potential security vulnerabilities.
- Incident Response and Monitoring: Developed and implemented incident response plans for PKI-related security incidents, and monitored certificate usage and potential security incidents.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Full Stack Developer(DF/Mulesoft/Java/Py)/Data Migration/Software Test Engineer
Responsibilities:

- Developed, maintained, and supported infrastructure solutions within multi-cloud environments, enabling seamless data and workflow integration between AWS, Azure, and specific Salesforce Clouds (e.g., Sales, Service, Marketing).
- Defined data requirements, structures, and transformation logic for complex data migration projects, resulting in streamlined data integration. Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.

- Collaborated with stakeholders to gather & document specific business and data quality requirements for the target system.
- Developed a suite of mapping documents to define data transformations between multiple source and target systems.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis.
- **Docker Images:** Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Deployed Artifactory for secure artifact control & incorporated Jira for project oversight and issue resolution throughout the CI/CD lifecycle.
- **Automated Certificate Lifecycle Management**: Developed and implemented automated certificate lifecycle management processes, including issuance, renewal, and revocation, to maintain a secure and compliant PKI infrastructure.
- Salesforce PKI integration:Integrated PKI with Salesforce Certificate and Key Management, to configure certificate-based authentication for Salesforce APIs, and to deploy certificates for secure communication.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Software Test Engineer/ Data Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Developed, maintained, and supported infrastructure solutions within multi-cloud environments, enabling seamless data and workflow integration between AWS, Azure, and specific Salesforce Clouds (e.g., Sales, Service, Marketing).
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /EDW
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Designed, developed, and deployed efficient and scalable data integration solutions for on-premises data pipelines using Enterprise Ab Initio's ETL tools (GDE, ADC, Express-IT, Control Center) and advanced techniques (Meta Programming, Co>Relate). Implemented robust data quality features within Ab Initio-based data pipelines. Performed database backups (SQL, MongoDB, DynamoDB).
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, AI Agents, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.3, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Azure Engineer/Data-AI Consultant – UpTech – Nuage -2/6/25- Interviewed on 4/6/25
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Azure Engineer/Data-AI Consultant/Software Engineer /Senior DevOps Engineer/Clouds/Linux Administrator/ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW /DBA-Japan(7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Azure Engineer/Data-AI Consultant/Software Engineer/Snr DevOps Engineer/Py/AI/ML/SP/D365 Developer
Responsibilities:

- Partnered closely with aged care regulatory bodies to design and build modern cloud solutions on the Azure stack, leveraging Azure Functions and App Services to automate critical business operations. This involved integrating OpenAI for intelligent document analysis and Cognitive Search for rapid retrieval of compliance data, all underpinned by the scalable data capabilities of Cosmos DB, significantly enhancing performance and smart automation.
- **Azure Functions**: Developed serverless APIs and event-driven processes using Azure Functions to automate real-time data ingestion and workflow orchestration, reducing processing time by 30%. E.g:
  public static async Task<IActionResult> Run([HttpTrigger(AuthorizationLevel.Function, "post")] HttpRequest req)

- **Azure App Services**: Engineered scalable web applications and RESTful APIs deployed on Azure App Services, enabling seamless integration with front-end platforms and external systems. E.g: services.AddControllers().AddNewtonsoftJson(options => { options.SerializerSettings.ReferenceLoopHandling = Newtonsoft.Json.ReferenceLoopHandling.Ignore; });
- **Azure OpenAI for AI-Driven Risk Assessment**: Implemented Azure OpenAI solutions for AI-driven risk assessment in regulatory compliance, extracting critical insights and identifying potential anomalies from large volumes of unstructured text data. E.g: response = openai_client.chat.:completions.create(model="gpt-4", messages=[{"role": "user", "content": "Analyze risk in audit report:..."}])
- **Azure Cognitive Search with Cosmos DB**: Designed and deployed high-performance search solutions using Azure Cognitive Search for rapid information retrieval from regulatory documents, leveraging Cosmos DB as a highly available and globally distributed data source..E. g: SearchClient searchClient = new SearchClient(serviceEndpoint, indexName, credential); SearchResults<MyDocument> results = searchClient.Search<MyDocument>("compliance breach");
- Designing and developing various APIs to connect internal and external data systems. These APIs may be built and managed using Azure services such as Azure Logic Apps, Azure Functions, Service Bus, & API Management.
- Proven Azure DevOps Engineer with a strong "Automation First" mindset and capability to review and uplift existing Azure infrastructure and deployment workflows, with a primary focus on identifying and implementing comprehensive automation solutions leveraging Terraform, Python/Bash, Azure DevOps, Docker, and AKS. E.g.: Terraform: resource "azurerm_kubernetes_cluster" "aks" { name = "my-aks" location = "australiaeast" resource_group_name = "rg-aks" }
- **Azure DevOps**: pipeline = azurerm.Pipeline(name="app-deploy", project=project.id, yaml_path="app-deploy.yaml")
- **Docker**: docker build -t my-app:latest . **Azure Kubernetes Service (AKS)**: kubectl apply -f deployment.yaml
- Integrated with Power Platform (Power Automate, Power Apps, Power BI) by designing & developing Custom Connectors for secure API interaction (via Azure API Management, Azure API for FHIR, Azure Container Apps, Azure Spring Apps), and enhanced automation with AI Builder. This included developing Power Apps Component Framework (PCF) controls for specific and enhanced visualization of fetched external CRM data within the model-driven application interface.
- Developed medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV & Azure AI API, enabling accurate, compliant, & personalized medical insights and decision support.
- Configured and utilized Power Platform Custom Connectors to establish secure data retrieval from Azure API Management (APIM) within Power Apps and used Power Automate to automate repetitive tasks across different applications & services.
- Developed AI APIs on a secure, isolated NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabelled data & supervised KNN within Python for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Developed Power Pages websites using Power Pages Design Studio to seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users
- Developing Agent App streamlining Aged Care Sales, using Google's newly announced Agent-to-Agent (A2A) protocol for inter-agent communication & Model Context Protocol (MCP) for seamless integration with external tools (web search APIs).
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI Agents (Ollama 4, Google NotebookLM) till MCP using Agent2Agent portocol) for data analysis and decision making.
- Designed and implemented AI-driven workflows using n8n to automate client intake, medical history triage, and wellness reporting for aged care service, integrating AI agents & LLMs to enhance operational efficiency & personalized aged care.
- Architected and implemented robust financial payment security solutions leveraging Azure's native security services, including Azure Firewalls, WAF, and DDoS Protection, alongside Microsoft Defender and Intune for endpoint protection.
- Proficient in applying Zero Trust principles via Azure IAM and Key Vault to secure identities and privileged access, while utilizing Microsoft Sentinel and SIEM/SOAR technologies for AI-powered anomaly detection, fraud prevention, and rapid incident response. Demonstrated alignment with PCI DSS, ISO 27001, and global financial security regulations across network, workstation, and cloud environment.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Azure Engineer/Data-AI Consultant/Snr DevOps Engineer/Py/AI/ML/D365/SP Developer /EDW
Responsibilities:

- Designed and implemented modern cloud solutions on the Azure stack in close collaboration with RBA financial services stakeholders, employing Azure Functions and App Services to automate essential financial workflows. The integration of OpenAI for intelligent document processing and Cognitive Search for quick retrieval of financial compliance data, supported by the scalability of Cosmos DB, dramatically enhanced operational efficiency and intelligent automation.
- **Azure Functions**: Implemented secure, serverless event processing for RBA financial data using Azure Functions. E.g.: [FunctionName("ProcessFinancialData")] public static async Task Run([QueueTrigger("rba-financial-queue", Connection = "StorageAccount")] string financialRecord)
- **Azure App Services**: Developed secure & scalable RESTful APIs on Azure App Services for RBA financial data exchange: builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme).AddMicrosoftIdentityWebApi(builder.Configuration.GetSection("AzureAd"));

- **Azure OpenAI for AI-Driven Compliance Analysis**: Leveraged Azure OpenAI to extract key insights and identify potential compliance risks from unstructured RBA financial documents. E.g.: extracted_insights = openai.Completion.create (engine="text-davinci-003", prompt="Summarize key compliance risks in the following RBA report: ...")
- Azure Cognitive Search for Financial Risk Intelligence: Built high-performance search functionality with Azure Cognitive Search to rapidly locate risk-related information within RBA financial documents indexed from Cosmos DB. E.g.: SearchResults riskFindings = searchClient.Search("market risk", new SearchOptions { Select = new[] { "title", "summary", "documentId" } });
- Developed, constructed, and implemented bespoke Power Apps (Canvas and Model-driven) applications, alongside automating workflows via Power Automate and establishing secure data structures and entities in Dataverse, further enhanced by the creation and upkeep of insightful Power BI dashboards and reports.
- Architected and implemented robust financial payment security solutions leveraging Azure's native security services, including Azure Firewalls, WAF, and DDoS Protection, alongside Microsoft Defender and Intune for endpoint protection.
- Proficient in applying Zero Trust principles via Azure IAM and Key Vault to secure identities and privileged access, while utilizing Microsoft Sentinel and SIEM/SOAR technologies for AI-powered anomaly detection, fraud prevention, and rapid incident response. Demonstrated alignment with PCI DSS, ISO 27001, and global financial security regulations across network, workstation, and cloud environ
- Designed, built, and deployed custom Power Apps (Canvas and Model-driven) applications while automating business processes using Power Automate and establishing robust data models and entities within the Dataverse, complemented by the development and maintenance of impactful Power BI dashboards and reports.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics 365/Power BI/Power Automate & Power Apps)
- Developed end-to-end Power BI paginated reports utilizing Report Builder's direct connectivity to diverse data sources like SQL Server and Azure Synapse (bypassing underlying models), implementing intricate calculations with nested row context and visually enhanced conditional formatting (including gradients), alongside the creation of insightful Power BI dashboards leveraging complex DAX formulas for advanced data manipulation.
- Architected and implemented diverse APIs to facilitate connectivity between internal & external data platforms, suing Azure services including Azure Functions, Logic Apps, Service Bus, & API Management for their construction and maintenance.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Laye**r: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Azure Engineer/Data-AI Consultant/Software Engineer/Snr DevOps Engineer/Py/ML/SP/365 Developer
Responsibilities:

- Azure Infrastructure & Automation: Leveraged Infrastructure as Code (IaC) principles using Python with ARM templates to automate the provisioning and configuration of Azure network resources. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration within Azure.
- API Development & Integration: Designed and implemented custom solutions using Power Apps and Microsoft Flow, leveraging Mulesoft Anypoint API for scalable integrations and automation. Developed and deployed Azure Functions and Logic Apps to establish connectivity with external APIs, managed and secured through Azure API Management (APIM).
- Machine Learning/Deep Learning Model Development & Deployment: Developed and maintained Machine Learning (ML) and Deep Learning (DL) models to enhance customer engagement and sales by efficiently manipulating and extracting insights from big data. Contributed to the adoption of MLOps practices.
- Microsoft Power Platform & AI Builder (Retain as forward-looking if applicable): Designed, developed, and implemented solutions using MS Power Platform (Power Automate, Power Apps, Power BI) and AI Builder, including intelligent automation leveraging form processing, object detection, and predictive models. Developed custom Dynamics 365 customizations and extensions using Power Automate and AI Chatbot.

- Cloud Security: Implemented secure landing zones for cloud workloads using Azure Blueprints. Leveraged analytical techniques to detect anomalies and prevent fraud. Experienced with Azure IAM and Key Vault.
- Data Management & Analysis: Developed custom components using Mulesoft Anypoint API to create interactive UI elements. Utilized KingswaySoft SSIS Integration Toolkit for data integration within Dynamics 365 environments. Optimized SQL queries for efficient data extraction and transformation.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps, Azure Pipelines
    2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting MLOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
  # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Azure Engineer/Data-ML Consultant/DevOps Engineer/Py/SP/D365 Developer/Clouds/Linux Admin.
Responsibilities:

- Developed and implemented data integration solutions, including ETL/ELT processes, within the Microsoft ecosystem.
- Designed data models, loading approaches, and source-to-target mapping documentation to facilitate efficient data flow.
- Analyzed diverse data sources, types, and formats to define requirements for data transformation, loading, and modeling.
- Utilized Python for custom ETL/API orchestration, demonstrating adaptability across different application environments.
- Contributed to the implementation of automation frameworks and CI/CD pipelines.
- Applied DevOps principles to automate build and deployment processes for online applications.
- Leveraged Python programming skills for data-related tasks and automation.
- Worked with various database systems, including SQL, Oracle, PostgreSQL, MySQL, DynamoDB, and MongoDB.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Designed, developed, and maintained automated ETL/ELT data pipelines and API orchestration using either:
    - **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.
    - **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.

- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Dynamics CRM Developer /DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Executed ELT/ETL processes and advanced analytics using R and Python for data enrichment and analysis.
- Designed DWH concepts from Bill Inmon to Ralph Kimball and build data models using Toad Data Modeler 6.x
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery.
- Designed and built cloud-based infrastructure solutions in Azure, optimizing performance and scalability.
- Utilized Azure DevOps (or similar CI/CD platform) for version control and build automation.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Developed OLAP Cubes either using SSAS or Cognos TM1 & visualize data either using Power BI, Tableau 10 or QlikView
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS
- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.

- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, OpenAI o1, GPT-3.5/4o, AI Agent, n8n ai agent, Azure OpenAI ML Services, Copilot, Google GEMINI Pro 1.5~2.5 NotebookLM; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Cloud Data Platform Engineer – Macqurie Bank – Randstad – 23/May/2025 – Interviewed by two Chinese guy
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Platform Engineer/Integration Solution Architect/Snr DevOps /Data Engineer/Java/.Net /Python Programmer / Linux Administrator /AI/ML/DL/BI /MuleSoft /SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Cloud Data Platform Engineer/Snr DevOps/Data Engineer/Linux Admin./Py/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- Architected and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for secure data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with AWS CloudWatch.
- Integrated M365 environments (O365, MS Teams, SharePoint 2022, Dynamics 365, Power App, Microsoft Flow) into the AWS Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Professionally leveraging AWS services (S3 for scalable storage, EC2 for compute, and managed container services like EKS and ECS Fargate for scalable application deployment), designs and implements robust big data pipelines using Spark, Presto, and Hive for analytics, orchestrated by Airflow for efficient workflow management.
- Expertise spans the entire SDLC, utilizing tools like Jira, Git, Jenkins, Bamboo, and Confluence within Agile and DevOps frameworks to automate development, deployment, and operational workflows, including the creation and configuration of EMR clusters for large-scale data processing within a Cloud-Native Data Lakehouse Architecture on AWS.
- Linux Server Provisioning: Leveraged Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) for automated EC2 instance creation, network configuration, package installation, and application deployment, often integrated into CI/CD pipelines via Jenkins or Bamboo.
- Leveraged Infrastructure as Code (IaC) principles with Terraform and Python (Boto3) to automate the provisioning and configuration of VPCs, subnets, and Network Security Groups (NSGs). Utilized AWS EKS and ECS Fargate for serverless container orchestration, enabling efficient and scalable deployment of containerized applications in Docker.
- Proven track record in administering Windows Server, with a focus on DNS, DHCP, Active Directory, and Group Policy.
- Provided technical support for critical business applications. Managed and optimized Azure, AWS and GCP infrastructures.
- Developed and implemented CI/CD pipelines using Azure Pipelines/AWS Code Pipeline/GCP Cloud Build, GitHub Actions and YAML scripting to automate the deployment of business applications, streamlining development & improving efficiency.
- Performing Big Data analysis by leveraging Spark for advanced processing and transformations, Presto for interactive querying across diverse datasets, and Hive for structured data warehousing, all orchestrated by Airflow for robust pipeline management within EMR clusters to support comprehensive Enterprise Data Warehouse (EDW) initiatives.
- Drives Big Data analysis and Enterprise Data Warehouse (EDW) initiatives by architecting and managing data pipelines that span Cloudera clusters and AWS environments, employing AWS Glue for complex transformations, AWS Redshift Spectrum for interactive data exploration, and AWS Redshift for high-performance warehousing, ensuring robust orchestration through AWS SDK or managed services
- Proficient in the creation and configuration of EMR clusters, with proven experience in complex Airflow upgrades, including the significant transition from v2.0 to v3.0.
- **Security**: Proficient in implementing security best practices, including Security Groups, Network ACLs, and AWS WAF or AWS Network Firewall, and VPN configurations (AWS Site-to-Site VPN/Client VPN).
- Design, develop, and maintain automated ETL/ELT data pipelines and API orchestration using either:
    - **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.

    - **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Design, develop, and implement solutions using MS Power Platform (Power Automate, Power Apps, Power BI) & AI Builder.
- Developed medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV & Azure AI API, enabling accurate, compliant, & personalized medical insights and decision support.
- Developed AI APIs on the NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabeled data and supervised KNN within R for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI Agents (Ollama 4, Google NotebookLM) till MCP using Agent2Agent protocol) for data analysis and decision making.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Cloud Data Platform Engineer/Snr DevOps/Data Engineer/Mulesoft/Python/AI/ML/SP/BI Developer/EDW
Responsibilities:

- Designed and implemented robust application and platform solutions using Mulesoft and cloud-native CI/CD tools (e.g., Azure DevOps, AWS DevOps, GCP Cloud Build/Cloud Deploy). Ensured alignment with RBA architectural standards to support critical organizational systems and services, integrating across AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Built and implemented Medallion Architecture with Data Lake in AWS:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting. kinesis_client = boto3.client('kinesis', region_name='ap-southeast-2')
- Leveraged Matillion and ERwin to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs, within both enterprise data warehouse (Inmon) and data mart (Kimball) architectures.
- Engineered seamless data and workflow integrations leveraging MuleSoft Anypoint Platform, including API development and hybrid deployment strategies, and integrated with AWS API Gateway and Azure IPaaS.
- Developed and optimized ETL/ELT data pipelines using AWS Glue, Python, SQL, and Apache Spark, migrating and transforming data between on-premises systems and cloud platforms, including Salesforce.
- Designed and implemented a secure hybrid cloud architecture integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- With deep expertise in AWS services—including S3 for scalable storage, EC2 for compute, and managed container platforms like EKS and ECS Fargate—designs and deploys resilient big data pipelines using Spark, Presto, and Hive for advanced analytics, orchestrated with Airflow to streamline workflow automation.
- Proficient in the full software development lifecycle, harnessing tools such as Jira, Git, Jenkins, Bamboo, and Confluence within Agile and DevOps environments to optimize development, deployment, and operational processes, including the configuration of EMR clusters for high-performance data processing within a Cloud-Native Medallion Architecture on AWS.
- Skilled in Linux server provisioning, leveraging Ansible playbooks (e.g., ec2_instance.yml, network_config.yml, package_install.yml, deploy_app.yml) to automate EC2 instance creation, network configuration, package management, and application deployment, seamlessly integrating these processes into CI/CD pipelines via Jenkins or Bamboo.
- Implemented robust CI/CD pipelines with Mulesoft to automate application delivery and reduce manual intervention. Leveraged Terraform to efficiently provision and manage cloud infrastructure resources (Azure, AWS, GCP) for Mulesoft Anypoint API deployments connecting to major financial institutions 24x7.
- Design, develop, and maintain automated ETL/ELT data pipelines and API orchestration using either:
    - **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.
    - **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime

- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless failover between AWS and Azure. This ensures resilience against outages.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Optimized data pipelines utilizing Airflow and Luigi, reducing processing time by 30% and improving resource utilization.
- Automated model versioning and rollbacks with MLflow, enabling seamless rollbacks and mitigating production issues: e.g. mlflow.sklearn.log_model(model, "linear_model") & rollback by client.transition_model_version_stage()
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Developed and implemented robust CI/CD pipelines using GitHub Actions to streamline Mulesoft application deployment. Leveraged YAML scripting and automation to enhance RBA development efficiency and reduce risk.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Cloud Platform Engineer/Snr DevOps /Data Engineer/Mulesoft/Python/ML/SharePoint/K2 Developer/EDW
Responsibilities:

- Cloud Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Strategic planning regarding optimizations for the improvement of the core infrastructure.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created an Amazon Redshift data warehouse (or AWS Lake Formation with Amazon Athena/Redshift Spectrum) in the AWS Management Console to provision storage and compute for transformed data.
- Analyzed requirements, planned tests, and collaborated effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- Utilized Terraform/CloudFormation (or similar) to automate the provisioning of cloud infrastructure for a PySpark-based data platform, ensuring consistent and reproducible deployments.
- Integrated AWS ECS service into the data platform CI/CD pipeline, automating containerized deployments and ensuring seamless integration with AWS infrastructure.
- Automated ETL/API orchestration using Python (Airflow) or Java (Spring Batch), implementing CI/CD pipelines for seamless deployment, monitoring, and infrastructure management.
- Automated AWS IaC pipeline creation and optimization using Python, SageMaker, CloudFormation, and boto3, emphasizing CI/CD best practices and infrastructure as code principles.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Designed and developed fully automated CI/CD pipelines using AWS DevOps, Git, Docker, GitHub Actions, and Jenkins
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Cloud Infrastructure Engineer/DevOps Engineer/Clouds/Linux Administrator
Responsibilities:

- Skilled Cloud Integration Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Cloud Infrastructure Engineering: AWS Expertise: Proficient in designing, implementing, and managing AWS infrastructure, including VPCs, subnets, Security Groups, Network ACLs, and AWS Network Firewall or AWS WAF (for web application firewall capabilities), AWS SSO, FortiGate Firewalls (as third-party appliances, which would run on EC2), FortiClient VPN with MFA (connecting to a VPN solution, likely AWS Client VPN or a self-managed VPN on EC2), and AWS IAM Identity Center (formerly AWS SSO)-based access control for AWS Secrets Manager or AWS Key Management Service (KMS).
- **Network Engineering**: Strong understanding of the interplay between CIDR, the TCP/IP stack, and routing protocols (IGPs: OSPF, RIP; EGP: BGP) in WAN and hybrid cloud environments.
- **SD-WAN**: Experience with SD-WAN solutions to optimize WAN performance and security.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban). Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all EML's data pipelines.
- Developed, reviewed, and maintained cloud support documentation and best practices
- Managed the integration of system solutions across various platforms and technologies, supporting DevOps and Cloud.
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Streamlined DevOps processes by implementing robust version control practices with Git, including branching strategies, code reviews, and merge requests. Utilized configuration management tools to consolidate settings and streamline infrastructure management, ensuring consistent provisioning.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Infrastructure Engineer/Snr Network/DevOps Engineer/SharPoint Developer
Responsibilities:

- Proven track record of designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow.
- Designed, developed and maintained automated CI/CD pipelines using platform-specific tools:
    1. Azure: Azure DevOps, Azure Pipeline
    2. AWS: AWS CodePipeline/CodeBuild/CodeDeploy
    3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to clients' specific needs.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Administrator (AD/SCCM/SCOM)/DBA
Responsibilities:

- Managed and maintained Azure AD, D365 CRM, SharePoint, and M365 for a large organization, supporting over 1000 users.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Implemented JIRA project management tool, streamlining issue tracking and collaboration across cross-functional teams.

- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Implemented advanced firewall configurations and intrusion prevention systems, safeguarding against cyber threats.
- Conducted regular network performance monitoring and optimization, enhancing network efficiency by 20%.
- Designed and managed network infrasctructure and security measures.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX A100s systems, NVIDIA Jetson Orin AGX 64GB Developer Kit ; OpenAI, ChatGPT-1.0~4o, Multi-GPTs, Auto-Agents, AI Agents, Azure Open AI ML Services, Copilot, Google GEMINI, NotebookLM, Perplexity, SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1

10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0

- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
- **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Power Automate Developer –Parramatta Gov- Mane Consulting – 18 May 2025 – No Resp. F.it.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Power Automate Developer/Software Engineer /Senior DevOps/Data Engineer/Clouds/Linux Administrator/ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW/DBA-Japan(7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Power Automate Developer/Snr DevOps/Data Engineer/.NET/Python/AI/ML/SP/D365 Developer
Responsibilities:

- Design, develop, and implement solutions using MS Power Platform (Power Automate, Power Apps, Power BI) & AI Builder.
- Designed and developed Power Apps Component Framework (PCF) controls to provide a specific and enhanced visualization of fetched external CRM data within the model-driven application interface.
- Automated business processes and workflows across departments, resulting in enhanced efficiency and service delivery.
- Designed, built, and deployed custom Power Apps (Canvas and Model-driven) applications while automating business processes using Power Automate and establishing robust data models and entities within the Dataverse, complemented by the development and maintenance of impactful Power BI dashboards and reports.
- Designed and implemented Power Pages websites that effectively presented and captured data from Dynamics 365 while integrating SharePoint Online for document sharing and leveraging virtual tables to expose SharePoint list data.
- Architecting and implementing intelligent automation solutions by leveraging AI Builder features such as form processing, object detection, and predictive models within Power Apps and Power Automate workflows.
- Integrated with Microsoft Power Platform (Power Automate, Power Apps, Power BI) by designing & developing Custom Connectors to securely access and interact with APIs exposed through Azure API Management or other services (including Azure API for FHIR and serverless compute like Azure Container Apps or Azure Spring Apps), and enhanced automation workflows by leveraging AI Builder features for intelligent data processing and analysis.
- Designing and developing various APIs to connect internal and external data systems. These APIs may be built and managed using Azure services such as Azure Logic Apps, Azure Functions, Service Bus, & API Management.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements to develop Dynamics 365 applications, using customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems. Liquid for templating & rendering dynamic content within web applications.
- Design and develop custom Dynamics 365 customizations and extensions to streamline business processes, using Common Data Service, Power Automate, and AI Chatbot (formerly Power Virtual Agents)
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools
- Developed medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV & Azure AI API, enabling accurate, compliant, & personalized medical insights and decision support.
- Gathered requirements through stakeholder collaboration to deliver tailored solutions, ensured the reliability of existing Power Automate flows and integrations through ongoing maintenance and support, and contributed to best practice development and documentation efforts.
- Configured and utilized Power Platform Custom Connectors to establish secure data retrieval from Azure API Management (APIM) within Power Apps and used Power Automate to automate repetitive tasks across different applications & services.

- Developed AI APIs on the NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabeled data and supervised KNN within R for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Developed Power Apps model-driven app capabilities to fetch and display real-time data from an external CRM system, enhancing Business Analyst access to critical information.
- Developed Power Pages websites using Power Pages design studio that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- Designed and deployed a comprehensive O365 framework, utilizing tools such as Teams for streamlined workflows, OneDrive and SharePoint Online for centralized information access, and Yammer/VIVA Engage to improve internal communication and knowledge sharing, resulting in significant gains in team productivity and collaborative efficiency.
- Developing Agent App streamlining Aged Care Sales, using Google's newly announced Agent-to-Agent (A2A) protocol for inter-agent communication & Model Context Protocol (MCP) for seamless integration with external tools (web search APIs).
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI Agents (Ollama 4, Google NotebookLM) till MCP using Agent2Agent portocol) for data analysis and decision making.
- Designed and implemented AI-driven workflows using n8n to automate client intake, medical history triage, and wellness reporting for aged care service, integrating AI agents & LLMs to enhance operational efficiency & personalized aged care.
- **Reduced development effort by 50% by leveraging OpenDevin** to eliminate the need for custom HTML, JavaScript, and canvas manipulation when customizing Dynamics 365, SharePoint 2022, & Power Apps using C#.NET Core in VS2022.
- Automated CI/CD pipelines using Azure DevOps, Git, Docker, Kubernetes, GitHub Actions, and Jenkins
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior Power Automate Developer/Snr DevOps/Data Engineer /Python/ML/SP/Power BI/D365 Developer
Responsibilities:

- Utilise MS Power Platform (Power Automate, Power Apps, Power BI) to architect, create, and deploy solutions.
- Engineered and built Power Apps Component Framework (PCF) controls for tailored and improved presentation of sourced external CRM data within the model-driven application environment. Streamlined organisational processes and workflows spanning various units, leading to greater productivity and improved operational effectiveness.
- Developed, constructed, and implemented bespoke Power Apps (Canvas and Model-driven) applications, alongside automating workflows via Power Automate and establishing secure data structures and entities in Dataverse, further enhanced by the creation and upkeep of insightful Power BI dashboards and reports.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power Apps)
- Architected and implemented diverse APIs to facilitate connectivity between internal & external data platforms, suing Azure services including Azure Functions, Logic Apps, Service Bus, & API Management for their construction and maintenance.
- Deployed the KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 to build Dynamics 365 applications through customisation and extension components, thereby optimising data movement and enabling integration with SAP HR, SAP FinOps, S/4 HANA systems, alongside utilizing Liquid for dynamic content templating and rendering in web applications.
- Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).
- Developed and deployed Azure Functions and Logic Apps to establish connectivity with external CRM APIs (primarily REST), enabling seamless data retrieval.
- Implemented Azure API Management (APIM) to secure, manage, and expose backend services as well-defined and governed APIs, ensuring controlled and reliable access.
- PowerBI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.

- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Layer**: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2017 to develop Dynamics CRM2016 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Microsoft Flow Developer/Snr DevOps/Data Engineer /AI/ML/SP/D365 /K2 Developer
Responsibilities:

- Software Integration Architect who successfully designed and implemented Mulesoft extensions to integrate the client's modern cloud-based HR platform with an on-site payroll system, automating employee onboarding & offboarding processes.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow to enhance usability and efficiency within Dynamics 365 & SharePoint 2016 environments. Leveraged Mulesoft Anypoint API for scalable integrations and automation, while also utilizing Microsoft Flow to streamline business processes.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.
- Designed and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps
    2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.

- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting AI and MLOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management. # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Microsoft Flow Developer/DevOps Engineer/Software Test Engineer/Clouds/Linux Administrator/DBA
Responsibilities:

- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2013 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM v8.0. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
- Collaborate with stakeholders to gather requirements and deliver tailored solutions.
- Maintain and support existing Power Automate flows and integrations.
- Contribute to the development of best practices and documentation.
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Microsoft Flow Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2011 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM. Leverage Visual Studio 2012 / VS Code, TFS, or Git for version control.
- Collaborate with stakeholders to gather requirements and deliver tailored solutions.
- Maintain and support existing Power Automate flows and integrations.

- Contribute to the development of best practices and documentation.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.

- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, OpenAI o1, GPT-3.5/4o, AI Agent, n8n ai agent, Azure OpenAI ML Services, Copilot, Google GEMINI Pro 1.5~2.5 NotebookLM; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Azure API Developer – DPHI – 6/May/2025 – Video Assessment done on 11/5/25- failed on 14/5/25
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Azure API Developer/Integration Solution Architect/Software Engineer/Snr DevOps/Data Engineer/SalesForce/ Java/.Net/Python Developer /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 Devloper /EDW –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Azure API Developer/Integration Solution Architect/Snr DevOps/Data Engineer /AI/ML/SP/D365 Developer
Responsibilities:

- Integration Solution Architects guide integration developers in the end-to-end lifecycle of integration solutions, from Azure API design and implementation through comprehensive testing for seamless application and service connectivity.
- Designing and overseeing the development of various APIs to connect internal and external data systems. These APIs may be built and managed using Azure services such as Azure Logic Apps, Azure Functions, Service Bus, & API Management.
- Facilitating integration with Microsoft Power Platform (Power Automate, Power Apps, Power BI) by designing & developing Custom Connectors or guiding developers on their use to securely access and interact with APIs exposed through Azure API Management or other services.These services include Azure API for Fast Healthcare Interoperability Resources(FHIR) and serverless compute options like Azure Container Apps or Azure Spring Appls if they host backend API logic.

- Developed medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV & Azure AI API, enabling accurate, compliant, & personalized medical insights and decision support.
- Providing architectural guidance for integrations using Mulesoft Anypoint API platform, ensuring alignment with the overall integration strategy and best practices, including considerations for hybrid scenarios involving Azure APIs. Focused on delivering solutions that drive business outcomes, enhance operational efficiency,& optimize processes through technology.
- Developing and delivering integration solutions using the Azure Integration Platform, including API Management, Azure Functions (on-demand code for API endpoints), Azure Logic Apps (workflow automation exposing APIs), Azure App Service (PaaS hosting for RESTful APIs), and Azure Event Hubs for high-throughput, real-time data streaming in API integrations.
- Capable of managing competing priorities while delivering high-quality, reliable solutions within project timelines ensured by:
- Implemented and managed comprehensive CICD automation processes, significantly enhancing release efficiency for Aged Care's Salesforce and MuleSoft platforms, while simultaneously facilitating the transition of legacy systems to a modern, MuleSoft Anypoint API-driven architecture.
- Developed and deployed scalable Azure Web Apps containerized with Docker and orchestrated using AKS, leveraging Azure Logic Apps for integration workflows with specific Azure APIs like Azure Blob Storage API and SQL Database API
- Configured and utilized Power Platform Custom Connectors to establish secure data retrieval from Azure API Management (APIM) within Power Apps to secure, manage, and expose backend services as well-defined and governed APIs, ensuring controlled and reliable access. Expertise in collaborating seamlessly with cross-functional teams for successful outcomes.
- Developed and deployed Azure Functions (serverless API endpoints) and Azure Logic Apps (automated API workflows) to establish connectivity with external CRM APIs (primarily REST APIs), enabling seamless data retrieval.
- Developed custom components using PCF Controls to build model-driven apps & reusable APIs, integrated into Dynamics 365 & Power Apps canvases. Automated data retrieval from backend SQL 2022 Database and SharePoint 2022 Document Lists to display frontend Dynamics 365 activity records, enhancing business workflow processes and customer services.
- Developed Dynamics 365 applications using KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements, employing customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems, with a focus on leveraging Azure APIs for a scalable and robust integration strategy. Liquid for templating & rendering dynamic content within web applications.
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools, and explored opportunities to integrate with Azure APIs for enhanced functionality.
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- Designed and deployed a comprehensive O365 framework, utilizing tools such as Teams for streamlined workflows, OneDrive and SharePoint Online for centralized information access, and Yammer/VIVA Engage to improve internal communication and knowledge sharing, resulting in significant gains in team productivity and collaborative efficiency.
- Strong problem-solving & analytical skills, with ability to identify issues & design effective technical solutions guaranted by:
- Developed complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI Agents (Ollama 4, Google NotebookLM) till MCP using Agent2Agent protocol) for data analysis and decision making.
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- Developed AI APIs on the NVIDIA Jetson Orin AGX 64GB Developer Kit, employing unsupervised learning for pattern discovery in unlabeled data and supervised KNN within R for reinforcement learning tasks. Expertly selected and implemented algorithms (KNN, CNN, RNN, ANN, GAN) based on data characteristics and interpretability requirements.
- Stakeholder & Client Management: Proven ability to collaborate effectively with diverse stakeholders and clients in complex environments, fostering strong working relationships to achieve shared goals.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Azure API Developer/Snr Integration Developer/Snr DevOps/Data Engineer/Java/.Net/AI/SP Dev.
Responsibilities:

- Design and develop Interactive Brokers (IB) OMS alike Trade WorkStation (TWS) applications either using Java 8/9/J2EE or IBPy (Python) for prime brokerage and prior trading within the FX, CFD, or fixed income, derivatives markets, leveraging Azure APIs for cloud integration. Develop high-performance, innovative code & software/system solutions for our customers.
- Skilled Full-Stack Engineer leveraging React/TypeScript for frontend, Java/APIs for backend development and expertise in AWS cloud. Proficient in integration tools like MuleSoft and J2EE for seamless data flow using RESTful web services, and capable of leveraging Azure APIs for integration.
- Developed financial-tailored LLMs (FinBERT, BloombergGPT, GPT-3 for Finance) and integrated ML/DL algorithms for financial data analysis with Pandas and Azure AI API, enabling accurate, compliant, and personalized financial insights and decision support. Provide regular status updates and reports on progress to ensure deadlines were met
- Providing architectural guidance for integrations using Mulesoft Anypoint API platform, ensuring alignment with the overall integration strategy and best practices, including considerations for hybrid scenarios involving Azure APIs. Focused on delivering solutions that drive business outcomes, enhance operational efficiency,& optimize processes through technology.

- Developing and delivering integration solutions using the Azure Integration Platform, including API Management, Azure Functions (on-demand code for API endpoints), Azure Logic Apps (workflow automation exposing APIs), Azure App Service (PaaS hosting for RESTful APIs), and Azure Event Hubs for high-throughput, real-time data streaming in API integrations.
- Capable of managing competing priorities while delivering high-quality, reliable solutions within project timelines ensured by:
- Implemented and managed comprehensive CICD automation processes, significantly enhancing release efficiency for RBA's Salesforce and MuleSoft platforms, while simultaneously facilitating the transition of legacy systems to a modern, MuleSoft Anypoint API-driven architecture.
- Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee), Azure API, and Azure IPaaS environments.
- Developed and deployed scalable Azure Web Apps containerized with Docker and orchestrated using AKS, leveraging Azure Logic Apps for integration workflows with specific Azure APIs like Azure Key Vault API and Azure Event Hubs API.
- Capable of managing competing priorities while delivering high-quality, reliable solutions within project timelines ensured by:
- Implemented and managed comprehensive CICD automation processes, significantly enhancing release efficiency for Aged Care's Salesforce and MuleSoft platforms, while simultaneously facilitating the transition of legacy systems to a modern, MuleSoft Anypoint API-driven architecture.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Prioritization & Execution: Proven ability to effectively manage competing priorities and consistently deliver high-quality, reliable solutions within established project timelines.
- Problem-Solving & Analysis: Strong problem-solving and analytical skills, adept at identifying root causes of issues and designing effective technical solutions.
- **Automated CI/CD pipelines** for data and machine learning models using AWS and Azure DevOps/Pipelines, Kustomize, Kubespray, Terraform for IaC, Docker for containerized applications, Kubernetes for scalability, Ansible, and GitLab CI/CD.
- **Technical Proficiency:** Developed and managed cloud-based applications using Azure Functions for serverless execution and Azure Kubernetes Service for container orchestration in microservice architecture.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power App)
- Developed serverless functionalities using Azure Functions with C# (.NET Core), Node.js, Python, Java, or PowerShell.
- **Specific Skill with API Management**: Designed and implemented serverless microservices using Azure Functions triggered by HTTP requests. Utilized Azure API Management to expose these functions as secure and manageable APIs.
- **Action-oriented with Achievements**: Implemented over 20 Azure Functions (C#, Node.js, etc.) for various backend tasks, improving API response times by 25%.
- Used Azure API Management to secure & expose these functions as public APIs, resulting in a 30% increase in API usage.
- Automated data exchange between disparate systems using MuleSoft Anypoint Studio 7 and Mule 4 Platform. Implemented MuleSoft DataWeave (or MEL) for data transformation, resulting in a 25% improvement in data processing speed.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless failover between AWS, Azure API, Azure, and GCP APIs. This ensures resilience against outages.
- Conduct testing, troubleshooting, and debugging of integration solutions to ensure they are functioning as expected in both development and production environments
- Undertake implementation activities to bring developed application "live" in the production environment and in accordance with agreed project deliverables and timelines
- Maintain clear documentation of the integration processes, solutions, and workflows enabling all team members to understand the solution

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Azure API Developer/Integration Solution Architect**/Sr DevOps/Data Engineer/.Net/Py/AI/SP/K2 Developer**
Responsibilities:

- Integration Solution Architects at the University of Canberra provide expert guidance to integration developers throughout the entire lifecycle of integration solutions. This includes the design and implementation of Azure APIs, ensuring comprehensive testing for seamless connectivity between applications and services.
- **API Design & Development for Data Connectivity**: Designed and oversaw the development of a variety of APIs to connect the University's internal and external data landscape, utilizing Azure services including Azure Logic Apps, Azure Functions, Service Bus, and API Management.
- **PaaS & Serverless API Integration on Azure**: Developed and delivered integration solutions utilizing Azure App Service (PaaS for RESTful APIs) and Azure Functions (serverless on-demand API endpoints), complemented by API Management, Azure Logic Apps for workflow automation, and Azure Event Hubs for real-time data streaming.

- Develop and Build: Use Azure DevOps pipelines, with GitHub Actions or GitLab CI/CD triggered by Git source code management, to automate builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk), including building & deploying necessary Azure API components.
- Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs (including AWS APIs exposed through API Gateway & Azure APIs within their IPaaS) that seamlessly integrate between AWS infrastructure and Azure IPaaS env.
- Developed University student-focused learning resources (lecture transcripts, study guides, past exam papers, interactive quizzes) and integrated ML/DL algorithms for student performance analysis with OpenCV & Azure AI API, enabling accurate, compliant, & personalized learning insights and academic support.
- Leveraged machine learning algorithms, including linear regression, logistic regression, decision trees, and random forests (using Scikit-learn and Keras), to build predictive models. These models were developed to support data-driven decision-making based on student input obtained through various APIs.
- **Technical Proficiency:** Designed and deployed scalable and reliable software applications on Microsoft Azure using Azure App Service and Virtual Machines. Implemented robust access controls and data protection for Azure environments
- Developed and deployed scalable Azure Web Apps containerized with Docker and orchestrated using AKS, leveraging Azure Logic Apps for integration workflows with specific Azure APIs like Azure Functions API & Azure API Management API
- Designed and implemented serverless microservices using Azure Functions triggered by HTTP requests. Utilized Azure API Management to expose these functions as secure and manageable APIs.
- Implemented over 20 Azure Functions (C#, Node.js, etc.) for various backend tasks, improving API response times by 25%.
- Used Azure API Management to secure & expose these functions as public APIs, resulting in a 30% increase in API usage.
- Delivering quality services and balancing competing demands to ensure project objectives were achieved.
- Developing broader knowledge of platforms and technologies to enable development of more complex solutions.
- Leveraged Microsoft Power Apps to design user interfaces and workflows for data management. Utilized Power Automate for automating data-related tasks, such as data collection, manipulation, and integration with various data sources, including triggering actions or accessing data within Azure API.
- Apply broad and thorough technical expertise to forecast needs, enhance the quality of the solution design and ensure optimal solutions are developed and delivered to meet customer needs.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Azure API Developer/Integration Developer /Software Test Engineer/DevOPs/Data Engineer/EDW/DBA
Responsibilities:

- Build and manage cloud-native applications and services, including App Services, Functions, Web Apps, and databases.
- Lead on-premise to cloud migration efforts with a focus on performance, security, and scalability.
- Develop robust ETL pipelines using Azure Data Factory and integrate data from various sources.
- Implement governance frameworks, usage policies, and security best practices across the cloud environment.
- Design, configure, and maintain databases in Azure including SQL, Cosmos DB, and Blob Storage.
- Support AI and machine learning workloads through deployment and operational readiness of Azure AI services.
- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- **Technical Proficiency:** Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Developed serverless functionalities using Azure Functions with C# (.NET Core), Node.js, Python, Java, or PowerShell.

- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Developed and deployed scalable Azure Web Apps containerized with Docker and orchestrated using AKS, leveraging Azure Logic Apps for integration workflows with specific Azure APIs like Azure Key Vault API and Azure Event Hubs API.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Azure API Developer/Systems Integration Developer/Software Test Engineer/.NET/SPoint Developer
Responsibilities:

- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Responsible for integration development process, ensuring adherence to standards and documentation of requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..

- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX A100s systems, NVIDIA Jetson Orin AGX 64GB Developer Kit ; OpenAI, ChatGPT-1.0~4o, Multi-GPTs, Auto-Agents, AI Agent, Azure Open AI ML Services, Copilot, Google GEMINI, NotebookLM, Perplexity, SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senoir Solutions Designer Developer – Sydney Water v2 – 2/4/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senoir Solutions Designer Developer/Integration Specialist /Software Engineer /Snr DevOps Engineer /Snr Data Engineer /Java/.Net/Python Developer /AI/ML/BI/TIBCO/MuleSoft/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Aust.(25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Solutions Designer Developer/Snr DevOps/Data Engineer/TIBCO/Java/Py/AI/ML/ML/SP/D365 Developer
Responsibilities:

- Integration Specialist specializing in designing and implementing robust integration frameworks to connect modern cloud-based software solutions with legacy on-premises applications. Leveraging cloud-native integration platforms like Azure Logic Apps, AWS Lambda, and TIBCO Cloud Integration, as well as traditional integration tools such as Tibco BusinessWorks and Tibco Mashery API Gateway, to automate approval processes for aged care approved providers.
- Designed, built, and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, Azure VNets, and TIBCO Cloud Integration Platform. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all four platforms. Leveraged CI/CD tools like Jenkins for automated provisioning and configuration of networks, and utilized AWS ECS Fargate for efficient and scalable container orchestration.
- Designed and implemented migration strategies from TIBCO BW5 to TIBCO BW Container Edition (Tibco BW CE Suite) 2.9.2, coupled with CI/CD pipeline development for automated deployments, improving overall system agility, with Kubernetes facilitating elastic scaling of TIBCO BW CE Suite 2.9.2 pipeline components. Competent in Kubernetes networking, Rancher security configurations, and architecting scalable microservice patterns for TIBCO integrations.
- Skilled in integration technologies, including TIBCO BW Container Edition (Tibco BW CE Suite), TIBCO Messaging (EMS and Kafka), Azure, SOA Suite, Axway API Gateway, Axway Managed File Transfer, cloud-native microservices, and cloud integration, with expertise in BPM, solution design, development, deployment, SOA governance, API management, and file transfer, including TIBCO BPM Enterprise Suite.
- Developed digital solutions (integrations, APIs, business process automations, cloud integration) on the TIBCO Container Platform, ensuring data quality, availability, security (with appropriate authentication, authorization, non-repudiation, and data encryption), and service stability.
- Proficient in TIBCO BW CE 2.92, enabling clients to build cloud-native applications with an API-first approach, designing and implementing robust APIs that serve as the foundation for their microservices architecture. Deploy these applications to container-based PaaS platforms such as Cloud Foundry™, Kubernetes, and OpenShift, or to similar Docker-supported cloud environments, leveraging the power of containerization to ensure rapid deployment, scalability, and portability.
- Designed and implemented integration solutions using either Mulesoft Anypoint Exchange/DataWeave or TIBCO BusinessWorks CE 2.92 to automate data exchange between disparate systems, improving data processing speed by 25%.
- Designed and implemented automated infrastructure using TIBCO BW Container Instances or AWS ECS Fargate, demonstrating strong understanding of architecture, cloud-native integration, containerization, patterns, and software development.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    - Azure: Azure DevOps, Azure Pipeline
    - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    - GCP: Cloud Build/Cloud Deploy
    - TIBOC: TIBCO Cloud Integration, TIBCO ActiveMatrix BW, TIBCO Cloud Mashery API, TIBCO BW CE Suite
- **Developed and deployed digital solutions (integrations, APIs, business process automations, and cloud integrations) either on the TIBCO Container Platform or on the Docker platform with Kubernetes.**
- **Develop Integration Flows**: Design flows in TIBCO BusinessWorks to act as intermediaries between systems.
- Utilize Connectors: Leverage pre-built connectors from TIBCO Cloud Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365, and Power Automate custom connectors.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, & implement security measures in TIBCO ActiveMatrix BW.
- Design and implement automated business process workflows primarily using TIBCO Workflow, in conjunction with Microsoft Power Automate, Power Apps, AWS/Azure DevOps, Python/PySpark, and Azure Data Factory for automated data ETL/ELT processes. Deploy applications to the TIBCO Cloud and utilize the TIBCO Messaging Suite, specifically EMS and Kafka, for real-time data streaming and performance monitoring.
- **Integration Solution Design with Azure DevOps CI/CD**: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (TIBCO BusinessWorks, Azure Logic Apps, or Hybrid).
- **Develop & Build**: Use Azure DevOps pipeline with source control (Git, BitBucket) to trigger builds for TIBCO BW CE, Azure IPaaS, BizTalk. Monitor and troubleshoot integration processes using ELK Stack, TOI Hawk, & Tibco BW Process Monitor.

- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI systems (PrivateGPT-X, Ollama 3.3, Google NotebookLM, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Design and implement a proactive fraud detection system, using risk control AI agents to identify & mitigate potential threats
- Technical support for aged care business applications covering SharePoint 2019/2022/Dynamics 365, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Solutions Designer Developer/Snr DevOps/Data Engineer/TIBCO/Java/AI/SP/BI/D365 Developer
Responsibilities:

- Integration Solutions Architect specializing in designing frameworks to integrate client's modern cloud-based software solutions with legacy applications (SAP BusinessObjects, SAP FinOps, and SAP S/4HANA), utilizing cloud-native integration with Azure Functions, AWS Step Functions, and Tibco Cloud Integration, Tibco BusinessWorks, and Tibco Mashery API Gateway for RBA financial services workflow automation.
- Designed, built, and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, Azure VNets, and TIBCO Cloud Integration Platform. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all four platforms. Leveraged infrastructure-as-code (IaC) tools like Ansible for automated provisioning and configuration of networks, and utilized **TIBCO Spotfire** for data visualization and analytics.
- Skilled in integration technologies, including TIBCO BW Container Edition (Tibco BW CE Suite), TIBCO Messaging (EMS and Kafka), Azure, SOA Suite, Axway API Gateway, Axway Managed File Transfer, cloud-native microservices, and cloud integration, with expertise in BPM, solution design, development, deployment, SOA governance, API management, and file transfer, including TIBCO BPM Enterprise Suite.
- Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using TIBCO BusinessWorks, TIBCO EMS, Java8/9/J2EE, or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, TIBCO BusinessWorks, TIBCO EMS, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or TIBCO API Gateway, with expertise in XML constructs, schemas, SOAP, and REST protocols.
- Proficient in creating DevOps pipelines & related frameworks, such as Agile, Lean, ITIL4, & DevSecOps, using tools like Git, Jenkins, Docker, Kubernetes, & Ansible, while leading solution design discussions at various levels within an organization.
- Leveraged either Mulesoft Anypoint Platform or TIBCO Cloud Integration's hybrid deployment capabilities to create flexible APIs that seamlessly connected GCP/AWS/Azure API platforms through their API Gateways and Azure IPaaS environments. By adopting a 'write-once, deploy-anywhere' approach, we eliminated the need for complex infrastructure management tools like Terraform, resulting in increased efficiency and reduced development time.
- Automated data workflows between diverse systems using Mulesoft Anypoint Exchange/DataWeave or TIBCO BW CE 7.x.
- Demonstrated expertise in enterprise-level integration technologies, including TIBCO BW Container Edition (Tibco BW CE Suite), TIBCO Messaging (EMS and Kafka), and TIBCO BPM Enterprise Suite, coupled with experience in Azure, SOA Suite, and Axway platforms (Axway API Gateway and Axway Managed File Transfer). Proven ability to design, develop, deploy, and govern complex integration solutions, including cloud-native microservices and API management, with a focus on optimizing data flow and business process automation.
- Designed and implemented robust, scalable infrastructure solutions using container orchestration platforms like TIBCO BusinessWorks Container Instances or AWS ECS Fargate. Successfully managed container provisioning, scaling, and deployment, ensuring high availability and performance. Contributed to improving operational efficiency and reducing manual overhead while maintaining strict security and compliance standards.
- Design, develop, and maintain automated data pipelines and API orchestration using either:
  - TIBCO Cloud Integration (CI) or TIBCO Data Virtualization (TDV) for data processing, integration, and API management, along with TIBCO Cloud Enterprise (TCE) for deployment and orchestration.
  - **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.
  - **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Implemented scalable digital solutions, including integrations, APIs, business process automations, and cloud integrations, utilizing cutting-edge technologies like TIBCO Container Platform or Docker with Kubernetes.
- **Integration Solution Design with Azure DevOps CI/CD**: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (TIBCO BusinessWorks, Azure Logic Apps, or Hybrid).

- **Develop & Build**: Use Azure DevOps pipeline with source control (Git, BitBucket) to trigger builds for TIBCO BW CE, Azure IPaaS, BizTalk. Monitor and troubleshoot integration processes using ELK Stack, TOI Hawk, & Tibco BW Process Monitor.
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., TIBCO CLI for TIBCO BusinessWorks, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Immersed experience adopting ISO27001, ACC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Solutions Designer Developer/Snr DevOps/Data Engineer/TIBCO/Java/Py/ML/SP/BI/D365/K2 Developer
Responsibilities:

- Integration Solutions Specialist focused on designing frameworks to integrate client's modern cloud-based software solutions with legacy applications (SAP HR, SAP FinOps, and SAP S/4HANA), utilizing cloud-native integration with Azure Functions, Azure Logic Apps, AWS API, and Tibco Cloud Integration, Tibco BusinessWorks, and Tibco Mashery API Gateway for streamlining student enrollment processes.
- Developed robust integration solutions using Mulesoft Anypoint Exchange/DataWeave or TIBCO BusinessWorks to streamline data exchange between various systems, boosting data processing efficiency by 25%.
- Implemented automated infrastructure solutions using TIBCO BusinessWorks Container Instances or Azure App Service for container orchestration. Effectively managed container provisioning, scaling, and deployment, streamlining operations and reducing manual overhead. Focused on application development and deployment, boosting efficiency and scalability.
- Skilled Full-Stack Engineer leveraging React/TypeScript for frontend, Java/APIs for backend development and expertise in AWS cloud. Proficient in integration tools like MuleSoft and J2EE for seamless data flow using RESTful web services.
- Designed and implemented automated infrastructure solutions using TIBCO BusinessWorks Container Instances or Azure Kubernetes Service(AKS). Successfully managed container orchestration, provisioning, scaling, and deployment, eliminating the need for manual infrastructure management. Focused on application development and deployment, resulting in improved efficiency and scalability.
- Developed and deployed digital solutions (integrations, APIs, business process automations, and cloud integrations) either on the TIBCO Container Platform or on the Docker platform with Kubernetes.
- Python or GoLang programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Overseeing the design and implementation of knowledge transfer programs to equip operational support teams with the necessary skills and expertise to maintain the solution's reliability and minimize operational risks.
- Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio

Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Solutions Designer Developer/DevOps Engineer/Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Integration Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins, BitBucket, and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH

Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, TFS) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET Core (Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, Mulesoft Studio 7 Mule 4, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc

- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, TIBCO Cloud Integration, TIBCO BW CE 5.x ~ CE2.92, Tibco Mashery API Gateway, Mulesoft Anypoint Exchange/DataWeave, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum.
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

of Seek site advertised Dynamics Technical Solution Architect/Developer position

Zhi Ming ZHANG - Australian Citizen File Ref: ZMZITILJP003
1303 / 29 Hunter St.| Parramatta 2150 | NSW Mobile: 0402-997-679 | mingjp@gmail.com

Hiring Manager
Snyk
Level 29/85 Castlereagh St, Sydney NSW 2000

Re: Application for Technical Support Engineer - Japanese Speaking

Dear Hiring Manager,

I am writing to express my enthusiastic interest in the Technical Support Engineer - Japanese Speaking position at Snyk, as advertised on Seek. With over 25 years of experience in Australia and 7.5 years in Japan in technical support, DevOps, cybersecurity, and software development, and my fluency in Japanese, I am confident I possess the skills and dedication to excel in this role and contribute significantly to your team.

My resume highlights my extensive experience in designing and implementing security solutions, particularly with Snyk vulnerability scanning and remediation, across diverse cloud environments. In my previous roles at Aged Care Quality Safety and Commission (ACQSC), Reserve Bank of Australia (RBA), and University of Canberra (UC), I successfully integrated Snyk into CI/CD pipelines, automated secure infrastructure provisioning, and resolved complex technical issues related to GitHub and cloud deployments.

I have a proven track record of providing exceptional technical support to both English and Japanese-speaking customers, utilizing various communication channels like email, chat, and GitHub issues. My ability to understand and reproduce complex issues, coupled with my proactive approach to problem-solving, aligns perfectly with the responsibilities outlined in your job description.

Moreover, my experience in implementing and managing IAM solutions, including CyberArk, Saviynt, and ForgeRock, and my deep understanding of security frameworks like ISO27001, ACSC Essential 8, and NIST, will enable me to effectively address the security needs of Snyk's customers. My proficiency in troubleshooting GitHub-related issues, as well as my experience with tools like Intercom and Jira, will allow me to streamline support processes and enhance customer satisfaction.

I am particularly drawn to Snyk’s mission to empower developers to build secure applications and your commitment to core values like Customer Centric and One Team. I am eager to leverage my technical expertise and language skills to contribute to Snyk's growth and success.

Thank you for your time and consideration. I look forward to the opportunity to discuss my qualifications further.

Sincerely,

Zhi Ming Zhang
mingjp2@gmail.com
Mobile: 0402 997 679

Application

of Seek site advertised Dynamics Technical Solution Architect/Developer position

Snyk 採用ご担当者様

件名:テクニカルサポートエンジニア(日本語対応)職 ご応募

貴社の Seek 掲載の求人情報を拝見し、テクニカルサポートエンジニア(日本語対応)職に大変魅力を感じ、応募させていただきます。張志明と申します。

オーストラリアで 25 年、日本で 7 年半のキャリアを通じ、テクニカルサポート、DevOps、サイバーセキュリティ、ソフトウェア開発に携わってまいりました。特に、Snyk のセキュリティリスクスキャンおよび修復に関するセキュリティソリューションの設計・実装において、豊富な経験を有しております。

Aged Care Quality Safety and Commission (ACQSC)、オーストラリア準備銀行 (RBA)、キャンベラ大学 (UC) では、CI/CD パイプラインへの Snyk 統合、セキュアなインフラストラクチャの自動プロビジョニング、GitHub およびクラウド展開に関連する複雑な技術的問題の解決に貢献してまいりました。

英語と日本語での顧客対応では、常に高い評価をいただいております。電子メール、チャット、GitHub Issues など、多様なコミュニケーションチャネルを通じて、顧客の課題に対し迅速かつ的確な解決策を提供してまいりました。複雑な問題の理解力と解決力は、貴社の求める人物像に合致するものと確信しております。

また、CyberArk、Saviynt、ForgeRock などの IAM ソリューションの実装・管理経験、ISO27001、ACSC Essential 8、NIST などのセキュリティフレームワークに関する深い理解は、貴社の顧客のセキュリティニーズに貢献できると確信しております。GitHub 関連のトラブルシューティング能力、Intercom や Jira などのツールに関する経験も、サポートプロセスの効率化と顧客満足度の向上に貢献できると自負しております。

貴社の開発者が安全なアプリケーションを構築できるよう支援するというミッション、そして顧客中心、ワンチームといったコアバリューに深く共感いたしました。私の技術的専門知識と語学力を活かし、貴社の成長に貢献したいと強く願っております。

つきましては、履歴書・職務経歴書を同封いたしましたので、ご査収いただけますと幸いです。お時間をいただき、ご検討いただきありがとうございます。 どうぞよろしくお願いいたします。

敬具

張 志明
mingjp2@gmail.com
Mobile: 0402 997 679

Technical Support Engineer-Japanese Speaking – Snyk – 31/3/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Snr Technical Support Engineer-Japanese Speaking/Snr DevOps/Data/Cyber Security Engineer/Snr Systems Engineer/ Linux/Unix Administrator/Java/.NET/Python/AI/ML/BI/SharePoint/D365 Developer/EDW -Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), **Sydney**
Gov. Health: Snr Technical Support Engineer-Japanese Speaking/Snr DevOps/CyberSecurity Engineer/Py/AI/SP/D365 Dev.
Responsibilities:

- Designed and implemented Snyk vulnerability scanning and remediation across hybrid cloud architectures, integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established secure code and dependency scanning pipelines using GitHub Actions & cloud-native integrations across all platforms, including configuration, development, design, and troubleshooting.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Teams/SharePoint 2019/Dynamics 365/Power BI/Power Automate & Power App).
- Provides technical support to both English & Japanese-speaking customers, answering & resolving their questions through email, chat & GitHub issues. Fluent in Japanese, enabling effective communication & support for our Japanese-speaking user base while building & maintaining trust with diverse clientele, balancing needs of enterprise clients & self-service users.
- Implemented Snyk Infrastructure as Code (IaC) scanning for Terraform, ARM, CloudFormation, and other configurations to proactively identify & remediate security vulnerabilities prior to cloud resource provisioning, enhancing deployment security.
- Integrated Snyk Container security scanning into CI/CD pipelines to identify and address vulnerabilities within Docker container images used for Kubernetes deployments before application deployment.
- Strengthened Identity and Privileged Access Management (IAM) maturity through the implementation and optimization of CyberArk, Saviynt, and ForgeRock for core IAM functions, complemented by Snyk's secure platform access management via IdP integration and analysis of IAM configurations in Infrastructure as Code.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- **Cloud Security with AI**: Leveraged AI tools to detect and prevent threats, manage vulnerabilities, and secure data.
- Design and implement a proactive fraud detection system, using risk control AI agents to identify & mitigate potential threats
- Demonstrated comprehensive understanding of Identity and Privileged Access Management (IAM) principles within the context of security platforms, including Snyk, and its interaction with tools like EntraID, CyberArk, and Saviynt Forgerock.
- **GitHub Repository Scanning Issues (Permissions)**: Diagnosed and resolved repository scan failures by meticulously reviewing and adjusting GitHub repository permissions, ensuring Snyk's access.
- **GitHub Pull Request Integration Problems (Checks Failing)**: Debugged and rectified webhook configurations between GitHub and Snyk, restoring accurate pull request checks and feedback.
- **GitHub Authentication and Authorization (Connection Problems)**: Successfully troubleshot and resolved OAuth and API key authentication issues, enabling seamless GitHub account connections to Snyk.
- **Snyk CLI and GitHub Actions (Workflow Errors)**: Identified and corrected environment variable and configuration errors within GitHub Actions workflows, ensuring accurate Snyk CLI execution.
- **Dependency Management & Vulnerability Remediation (Fixing Vulnerabilities)**: Provided clear, step-by-step guidance on vulnerability remediation, including dependency updates & configuration changes, minimizing application breakage.
- Communicates Effectively: Provides clear and concise technical explanations to diverse audiences, both verbally and in writing, to resolve customer issues.
- **Troubleshoots Complex Technical Problems**: Analyzes, reproduces, and resolves customer-reported issues, demonstrating a logical and analytical approach.
- **Demonstrates Technical Aptitude**: Proficient in utilizing source control tools like GitHub and eager to expand technical expertise through codebase exploration.
- **Diagnoses GitHub Actions and CI/CD Issues**: Identifies and resolves environment variable and configuration errors in GitHub Actions workflows to ensure successful Snyk CLI execution.
- Collaborates with Technical Teams: Works closely with engineering and QA teams to escalate and resolve complex technical challenges.
- **Develops Customer Support Resources**: Creates and maintains customer-facing documentation and FAQs, and contributes to improving support processes.

- Enhanced troubleshooting capabilities by integrating Slack with remote access tools such as CoScreen or TeamViewer. This allows for direct remote control during support sessions, significantly reducing resolution times.
- **Documentation and Project Management**: Utilized Intercom for detailed recording of troubleshooting sessions & customer interaction history. Employed Jira for structured documentation of issues and management of related support projects.
- Automated secure Linux server provisioning on Azure Virtual Machines (VMs) using Ansible Playbooks (azure_vm_instance.yml, network_config.yml, install_packages.yml, deploy_app.yml) and integrated Snyk to identify and address vulnerabilities in packages and configurations during the provisioning process.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Technical Support Engineer-Japanese Speaking/Snr DevOps/CyberSecurity Engineer/Py/BI Dev
Responsibilities:

- Engineered and deployed Snyk vulnerability scanning and remediation across hybrid cloud environments encompassing AWS VPCs, GCP Cloud VPCs, and Azure VNets, enhancing the security posture of interconnected systems. Established automated and secure code and dependency scanning pipelines leveraging GitHub Actions and cloud-native integrations across the SDLC, including configuration, development, design, and troubleshooting, to proactively manage risks.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Integrated M365 environments (O365, MS Teams, SharePoint 2016, Dynamics 365, Power App, Microsoft Flow) into the Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- Provided comprehensive technical support to both English and Japanese-speaking users, efficiently addressing and resolving their inquiries via email, chat, and GitHub issues. Leveraged fluency in Japanese to ensure clear and effective communication, building and maintaining trust with a diverse user base, including both enterprise clients and self-service users, while adhering to professional standards.
- Implemented Snyk Infrastructure as Code (IaC) scanning for Terraform, ARM, CloudFormation, and other infrastructure configurations to proactively identify and remediate potential security vulnerabilities prior to cloud resource provisioning, contributing to robust deployment security.
- Integrated Snyk Container security scanning into CI/CD pipelines to identify and mitigate vulnerabilities within Docker container images utilized for Kubernetes deployments prior to application deployment, enhancing the security of containerised applications.
- Contributed to a strengthened Identity and Privileged Access Management (IAM) framework through the implementation and optimization of CyberArk, Saviynt, & ForgeRock for core IAM functionalities, complemented by managing secure access to the Snyk platform via Identity Provider (IdP) integration and analysing IAM configurations within Infrastructure as Code.
- Implemented a robust zero-trust security framework using EntraID, CyberArk PSM10.8, Saviynt Forgerock, and Snyk to enhance identity and privileged access management and secure applications at RBA. This proactive approach significantly reduced the risk of unauthorized access and ensured compliance with industry regulations.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Diagnosed and resolved GitHub repository scanning issues related to permissions, ensuring Snyk's continuous and effective security analysis.
- Debugged and rectified webhook configurations between GitHub and Snyk to resolve pull request integration problems, ensuring timely security feedback within the development workflow.
- Identified and corrected environment variable and configuration errors within GitHub Actions workflows to ensure the accurate execution of Snyk CLI for automated security checks.
- Provided clear and actionable guidance on dependency management and vulnerability remediation, including recommending dependency updates and configuration changes to minimize application instability.
- Demonstrates technical proficiency in utilizing source control tools like GitHub and actively seeks to expand technical knowledge through codebase exploration.
- Identifies and resolves environment variable and configuration errors in GitHub Actions workflows to ensure the successful execution of Snyk CLI for continuous integration and security checks.
- Enhanced troubleshooting efficiency by integrating Slack with remote access tools such as CoScreen or TeamViewer, enabling direct remote control during support sessions and reducing resolution times.
- Automated the secure provisioning of Linux servers on Google Compute Engine (GCE) virtual machines using Puppet manifests (defining virtual machine creation, network configuration, package installation, and application deployment), integrating Snyk to proactively identify and address vulnerabilities in packages and configurations during the provisioning process, ensuring a secure baseline.
- **Cloud Security with AI**: Leveraged AI tools to detect and prevent threats, manage vulnerabilities, and secure data. Develop financial security-tailored LLMs (e.g., BLOOM-Fin, GPT-3-Finance) and integrated ML/DL algorithms for financial data analysis and risk assessment, enabling accurate, compliant, and personalized financial insights and decision support.

- **Network Security via SIEM:** Hybrid Cloud Security with Netskope SASE: Deploy Netskope Service Edge in data centers and cloud environments. Configure SASE functionalities including CASB, SWG, ZTNA, and DLP. Establish secure connections between on-premises and cloud (e.g., ExpressRoute, Direct Connect).
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- **Industry Compliance** (RBA financial market services): Implemented 5 key security controls aligned with ACSC Essential 8 (e.g., Application Control, MFA), ISM 37 securijectty controls (physical access controls with mandatory security clearances), & relevant NIST CSF categories (IAM, Protective Actions) to address RBA's financial sector risks & regulations.
- **Compliance Adherence**: Ensured compliance with ISO27001/2's 114 security controls and NIST800-53 standards.
- **Utilized various security tools:** Splunk, McAfee, Secret Server, CrowdStrike, Mimecast, Qualys, Tenable AD, Fortinet.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Technical Support Engineer-Japanese Speaking/Snr DevOps/CyberSecurity Engineer/Py/SP/K2 Dev.
Responsibilities:

- Designed and implemented Snyk vulnerability scanning and remediation strategies across hybrid cloud environments, integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets to enhance overall system security. Established secure code and dependency scanning pipelines using GitHub Actions and cloud-native integrations across various stages, including configuration, development, design, and troubleshooting.
- Designed and implemented secure landing zones for cloud workloads utilizing Azure Blueprints, with a focus on Windows-based systems within M365 environments (O365/MS Teams/SharePoint 2013/Dynamics 365/Power BI/Power Automate & Power App), establishing a secure and well-governed cloud foundation.
- Delivered technical support to a diverse user base of both English and Japanese speaking students, effectively answering and resolving their questions through various channels including email, chat, and GitHub issues. Utilized fluency in Japanese to facilitate clear communication and provide tailored support to Japanese-speaking customers, fostering trust and positive interactions with students and potentially university staff or project collaborators.
- Implemented Snyk Infrastructure as Code (IaC) scanning for Terraform, ARM, CloudFormation, and other configurations to proactively identify and remediate potential security vulnerabilities prior to cloud resource provisioning, contributing to improved deployment security and efficiency.
- Integrated Snyk Container security scanning into CI/CD pipelines to identify and address vulnerabilities within Docker container images utilized for Kubernetes deployments prior to application deployment, enhancing the security of containerised applications and deployment processes.
- Strengthened Identity and Privileged Access Management (IAM) maturity through the implementation and optimization of CyberArk, Saviynt, and ForgeRock for core IAM functionalities, complemented by the secure management of Snyk platform access via Identity Provider (IdP) integration and the analysis of IAM configurations within Infrastructure as Code.
- Applied knowledge and practical experience in adopting security frameworks and standards including ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS, and PCI-DSS.
- Contributed to the design and implementation of a proactive fraud detection system, utilising AI-driven risk control agents to identify and mitigate potential threats effectively.
- Demonstrated a comprehensive understanding of Identity and Privileged Access Management (IAM) principles within security platforms, including Snyk, and its interaction with tools such as EntraID, CyberArk, and Saviynt Forgerock.
- Diagnosed and resolved GitHub repository scanning issues related to permissions, ensuring the continuous and effective operation of Snyk security analysis.
- Debugged and rectified webhook configurations between GitHub and Snyk to resolve pull request integration problems, ensuring timely and accurate security feedback within the development lifecycle.
- Successfully troubleshooted and resolved OAuth and API key authentication challenges, maintaining seamless and secure GitHub account connections to Snyk.
- Identified and corrected environment variable and configuration errors within GitHub Actions workflows to ensure the accurate and reliable execution of Snyk CLI for automated security assessments.
- Provided clear and actionable guidance on dependency management and vulnerability remediation, including recommending and implementing dependency updates and configuration changes to minimize application vulnerabilities.
- Communicates technical information clearly and concisely to diverse audiences through both verbal and written means, facilitating effective problem resolution and knowledge sharing.
- Analyzes, reproduces, and resolves complex technical problems by employing a logical and analytical approach, contributing to efficient and effective technical support.
- Demonstrates technical aptitude through proficiency in utilizing source control tools such as GitHub and a proactive approach to expanding technical expertise through codebase exploration and learning.
- Identifies and resolves environment variable and configuration errors in GitHub Actions workflows to ensure the successful execution of Snyk CLI for continuous integration and security validation.
- Contributes to the development of customer support resources by creating and maintaining clear and helpful documentation and FAQs, and actively participates in improving support processes and knowledge management.

- Enhanced troubleshooting capabilities by integrating Slack with remote access tools such as CoScreen or TeamViewer, enabling direct remote assistance during support sessions and significantly improving issue resolution times.
- Utilized Intercom for detailed record-keeping of troubleshooting sessions and customer interaction history, and employed Jira for structured documentation of issues and efficient management of support-related projects.
- Provided technical support to Japanese-speaking customers, demonstrating strong communication skills and cultural sensitivity while building and maintaining trust with a diverse customer base, effectively balancing the needs of enterprise and self-service users.
- Automated the secure provisioning of Linux servers on Amazon Elastic Compute Cloud (EC2) instances using Puppet manifests (defining instance creation, network configuration, package installation, and application deployment), integrating Snyk to proactively identify and address vulnerabilities in packages and configurations during the provisioning process, ensuring a secure baseline.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior Technical Support Engineer/DevOps Engineer/Software Test Engineer/SharePoint Consultant
Responsibilities:

- **Designed and Implemented Robust Security Architectures**: Network security (SIEM, Netskope), DevSecOps practices, secure AWS landing zones, multi-cloud development pipeline, secure cloud infrastructure.
- **Security Architecture Enhancement**: Collaborated on enhancing security architectures, conducted threat intelligence analysis, responded to incidents swiftly, reviewed code for vulnerabilities, and ensured compliance with standards.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, SyCoverity, SonarQube, Twistlock) and IaC security.
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to workers compensation insurance and claims for schemes across Australia used for accessing sensitive data, systems, or applications related to claims processing, fraud investigation, or financial management. This automation process will ensure efficient and secure access management while ensuring strict compliance with data privacy regulations.
- #Determine privileged account assignments:account_assignments= determine_account_assignments(cv_expression_data)
- Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs of work.
- **Threat Mitigation**: Monitored and mitigated threats using tools like Wireshark and Metasploit.
- **Incident Response Planning**: Developed incident response playbooks for cloud-based services.
- **DevSecOps Integration**: Integrated CyberArk PSM security into GitHub Actions or GitLab CI/CD pipelines.
- Scrutinized code bases to eliminate common coding vulnerabilities and ensure secure application development and deployment. Deployed & managed advanced security tools to systematically identify & resolve security issues across systems & applications.Ensured adherence to industry-specific compliance standards & regulations in all security initiatives.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Utilized Splunk leading platform for log management, analysis, and security information and event management (SIEM), perform real-time monitoring, and conduct forensic investigations, including IDS, IPS, IAM and related.
- Configure McAfee and McAfee VPN to encrypts users' data and protects it from potential snoopers, ensuring that users' online activities and personal information remain secure even on untrusted networks.
- Setup and configured Tenable, Tenable.sc (formerly SecurityCenter) vulnerability management platform that helps organizations identify and prioritize security vulnerabilities across their infrastructure.
- Perform day-to-day monitoring and triaging critical alerts and coordinate with technical teams for incident response.
- Triage phishing emails, taking appropriate containment and remediation actions to prevent security incidents
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Technical Support Engineer/Cloud Engineer/DevOps/Data Engineer/SharePt Consultant
Responsibilities:

- **Security Architecture Enhancement**: Collaborated with cross-functional teams to enhance security architectures for financial investment projects.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within Azure cloud architecture.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- **Security Testing**: Performed penetration testing and vulnerability scanning.
- **Secure SDLC Implementation**: Advised development teams on secure coding practices.
- **NIST Best Practices**: Followed NIST guidelines for secure software development.
- Collaborating with IT Security for the Standard Operating Environment (SOE) security hardening in line with PSPF, ISM, ACSC, CIS and DTA Protected pattern guidelines.Stayed updated with industry security trends and emerging threats.

- Leveraged McAfee, Splunk, and Secret Server to enhance the effectiveness of Cyber Security Management programs.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Technical Support Engineer-Japanese Speaking/Cloud Security Engineer/Administrator /DBA
Responsibilities:

- **Cloud Security Design**: Designed secure cloud infrastructure, including IAM policies, network segmentation, and encryption.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within Azure cloud architecture.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- Configured a Netskope Cloud Private Access (NCPA) tenant and established secure connections between Azure and AWS environments to Netskope using virtual Private IPs or peering connections.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Coverity, SonarQube, Twistlock) and IaC security.
- **Network Security Enhancement**: Implemented network segmentation and intrusion detection systems.
- **Threat Hunting**: Proactively searched for vulnerabilities and potential threats.
- **Incident Handling**: Responded to security incidents, ensuring minimal impact.
- **Compliance Audits**: Conducted internal audits to meet industry standards.
- **Risk Assessment**: Evaluated emerging technologies for relevance and risk.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager/Technical Support Engineer-Japanese Speaking/Senior Systems Engineer
Responsibilities:

- Supported ongoing operational Cyber Security Management programs of work, contributing to a strong security framework.
- Managed hardware and server administration, ensuring reliable infrastructure performance.
- Oversaw AD, DFS, ADFS, AAD, DNS, and DHCP services, maintaining their stability and security.
- Provided level 3 reactive support for incident/service request management, monitoring, and event management.
- Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Technical Support Engineer-Japanese Speaking/Software Engineer/BI Developer /Data Engineer
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

Zhi Ming ZHANG - Australian Citizen File Ref: ZMZITILJP003
1303 / 29 Hunter St.| Parramatta 2150 | NSW Mobile: 0402-997-679 | mingjp@gmail.com

- **AI, ML, DL, BI IDE**: OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Reflection, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI, MakerSuite; SQL Server Management Studio/BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior DevOps Engineer/Full Stack Engineer – NDIA – Click IT ACT – 12/3/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Full Stack Engineer /Integration Solution Architect /Software Engineer/ Python/R Developer/ Salesforce/Mulesoft /AI/ML/BI/SharePoint/Dynamics 365 Developer /EDW/DBA –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr DevOps/Full Stack Engineer(SF/MuleSoft/Python/AI)/Snr Data Engineer/Salesforce Support Engineer/EDW
Responsibilities:

- Develop, maintain, and support infrastructure solutions using Microsoft technologies, specializing in data and workflow integration across AWS, Azure, and Salesforce Clouds (e.g., Sales, Service, Marketing).
- Designed and built highly available, secure & scalable hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Automated the deployment and management of secure infrastructure across hybrid cloud environments (AWS & Azure), integrating Postgres/MySQL databases with HashiCorp Vault for secure credential injection.
- Leveraged AWS CloudFormation for building scalable AWS Landing Zones. Implemented PKI for secure communication, and used Cloudflare for optimized network traffic. Federated Amazon Redshift access with Azure EntraID, SSO and MFA.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA, managing sensitive credentials and secrets using HashiCorp Vault.
- Architect, build, deploy, and provide support for the ACQSC's Cloud & Data Infrastructure. Work items are to be prioritized and managed via a work queue. Ensure solutions are compliant with integration best practices.
- Ensure business goals are achieved by engaging directly with business consumers. Being a vital part of our agile project, collaborating with stakeholders to define requirements and prioritize work.
- Demonstrated a high degree of autonomy in resolving complex technical challenges and managing critical infrastructure issues, applying sound judgment and decision-making to deliver senior-level technical guidance and support for cloud and data infrastructure initiatives.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure: Azure DevOps, Azure Pipelines**
  - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
  - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to your specific needs. Or

- Implemented robust data pipelines within CI/CD workflows using Kafka and Spark for efficient data processing, integrated with containerized applications deployed via Docker and Kubernetes. Utilized Artifactory for secure artifact management and integrated Jira for project tracking and issue management throughout the CI/CD pipeline.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Effectively managed stakeholder relationships, both internal and external, and served as a representative and negotiator for the ACSQS in diverse settings to advocate for its strategic interests.
- Developed and implemented ETL pipelines using AWS Glue, Python, and SQL to migrate Aged Care Approved Providers' data into Salesforce, supporting a critical aged care sector transformation.
- Enhanced AWS infrastructure automation through AI-driven CI/CD pipelines, utilizing SageMaker and CloudFormation to predict and optimize deployments. Implemented Python based automation that reduced repetitive tasks, and increased data analysis efficiency. This improved the overall efficiency of the development process.
- Developed & maintained CICD automation with GitLab CICD for ACQSC's Salesforce & MuleSoft, enabling faster releases. Utilized Git version control & integrated legacy systems with AWS, Azure & Salesforce cloud via MuleSoft Anypoint APIs.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., dfx force:source:pull) from Salesforce organizations.
- **Automated Deployments with SFDX**: Minimized manual errors and accelerated releases. Enhanced Code Quality: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.Independently architecting a multi-region, fault-tolerant Kubernetes cluster to support critical applications.
- Provided expert-level Salesforce support by investigating and resolving complex issues within OmniStudio components (FlexCards, OmniScripts, DataRaptors, Integration Procedures), performing comprehensive system administration (user management, application metrics monitoring, configuration and data integrity), managing deployments through a DevOps framework, and conducting thorough log analysis to diagnose and resolve system anomalies.
- **Multi-Cloud PKI Implementation**: Architected and implemented a comprehensive PKI solution across ACQSC's Azure, AWS, and Salesforce cloud environments, ensuring secure authentication and encryption for critical applications and APIs.
- Compliance and Security Audits: Conducted regular security audits and vulnerability assessments of the PKI infrastructure, ensuring compliance with industry standards and ACQSC's security requirements.
- Automated PKI deployment and configuration: Utilized Infrastructure as Code, to automate the deployment and configuration of PKI infrastructure across ACQSC's cloud enviroment, reducing manual errors, and increasing security.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr DevOps/Full Stack Developer(SF/MuleSoft/AI)/Snr Data Engineer/Salesforce Support Engineer/EDW
Responsibilities:

- Developed, maintained, and supported infrastructure solutions, demonstrating strong integration capabilities across AWS, Azure, and Salesforce Clouds (Sales, Service, Marketing), facilitating seamless data and workflow management.
- Engineered and deployed resilient, fortified, & adaptable multi-cloud architectures, interconnecting AWS VPCs, GCP Cloud VPCs & Azure VNets. Established secured data transmission pathways via Cloud VPN Gateways across all three platforms.
- Streamlined the deployment and governance of protected infrastructure within combined cloud landscapes (AWS & Azure), incorporating Postgres/MySQL databases with HashiCorp Vault for safeguarded credential integration.
- Designed, developed, implemented, and maintained the RBA's Cloud & Data Infrastructure. Tasks are to be prioritized and organized through a managed task list. Confirm that solutions adhere to established integration standards.
- Guaranteed that organizational objectives are met by direct interaction with business clients. Function as an essential member of our agile project team, cooperating with stakeholders to determine specifications and establish priorities.
- Exhibited significant independence in addressing intricate technical problems and overseeing essential infrastructure matters, employing sound reasoning and decision-making to furnish expert-level technical consultation and assistance for cloud and data infrastructure projects.
- Established resilient data processing pathways within CI/CD procedures, employing Kafka and Spark for streamlined data manipulation, seamlessly integrated with containerized applications deployed via Docker & Kubernetes.
- Automated cloud infrastructure deployment using Terraform and Packer for secure, consistent virtual machine deployments. Employed Ansible for post-deployment configuration administration and deployments on pre-configured images, complemented by integrated security assessments.
- Utilized AWS Glue Studio for data transformation, Apache Spark for data manipulation, AWS S3 for data repository, AWS Redshift for data storage, and MuleSoft Anypoint Platform for system connectivity, alongside proficiency in established technologies such as Teradata, Hadoop, Tableau, and SAS, and AWS-specific services like EMR and Redshift.
- Established and managed continuous integration/continuous delivery automation using GitLab CICD for RBA's Salesforce and MuleSoft platforms, facilitating accelerated release cycles. Deployed Git version control and connected heritage systems with AWS, Azure, and Salesforce cloud platforms through MuleSoft Anypoint APIs.
- Implemented Python-based automation that minimized recurring operations and amplified data analytical effectiveness. This enhanced the general productivity of the development lifecycle.

- Established and managed continuous integration/continuous delivery automation using GitLab CICD for RBA's Salesforce and MuleSoft platforms, facilitating accelerated release cycles.
- Deployed Git version control and connected heritage systems with AWS, Azure, and Salesforce cloud platforms through MuleSoft Anypoint APIs.
- Automated Salesforce Development Workflow: Utilized Salesforce DX tools (dfx force:source:pull) to retrieve and manage code (metadata) from development organizations. Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using dfx force:source:deploy.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., dfx force:source:pull) from Salesforce organizations.
- **Automated Deployments with SFDX**: Minimized manual errors and accelerated releases.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.Independently architecting a multi-region, fault-tolerant Kubernetes cluster to support critical applications.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless failover between AWS and Azure. This ensures resilience against outages.
- Design, develop, and maintain automated ETL pipeline/API orchestration using either:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **MuleSoft API Security with PKI**: Integrated PKI with MuleSoft Anypoint Platform to secure API communication, implementing certificate-based authentication and TLS/SSL encryption for seamless data exchange between systems.
- Secure Key Management: Established secure key management practices, utilizing Azure Key Vault and AWS Certificate Manager to protect private keys and ensure compliance with security policies.
- Hybrid Cloud Security Expertise: Demonstrated expertise in securing hybrid cloud environments by implementing a robust PKI strategy that supports seamless integration and secure communication across diverse platforms.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr DevOps/Full Stack Developer(SF/MuleSoft/AI)/Snr Data Engineer /Salesforce Support Engineer/EDW
Responsibilities:

- Expert in architecting and implementing complex cloud integration and data solutions within multi-cloud environments (AWS, Azure, Salesforce). Designed and deployed robust CI/CD pipelines using GitLab CICD, Terraform, Packer, and Ansible, automating infrastructure and application deployments across platforms.
- Built and implemented Medallion Architecture with Data Lake in AWS:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting. kinesis_client = boto3.client('kinesis', region_name='ap-southeast-2')
- Leveraged Matillion and ERwin to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs, within both enterprise data warehouse (Inmon) and data mart (Kimball) architectures.
- Engineered seamless data and workflow integrations leveraging MuleSoft Anypoint Platform, including API development and hybrid deployment strategies, and integrated with AWS API Gateway and Azure IPaaS.
- Developed and optimized ETL/ELT data pipelines using AWS Glue, Python, SQL, and Apache Spark, migrating and transforming data between on-premises systems and cloud platforms, including Salesforce.
- Implemented secure artifact control and project oversight using Artifactory and Jira.
- Proficient in Salesforce DX, automating development workflows and ensuring code quality through unit testing (Jest). Architected fault-tolerant Kubernetes clusters and containerized applications using Docker and Kubernetes.
- Skilled in data modeling (SPARX), data warehousing (AWS Redshift), and data analytics (SageMaker, Python, R).
- Implemented and managed secure certificate lifecycle management (PKI) and integrated it with Salesforce.
- Expert in troubleshooting and optimizing Salesforce OmniStudio components, ensuring data accuracy and compliance through robust validation rules.

- Demonstrated strong stakeholder management and communication skills, representing the organization in strategic forums. Leveraged intelligence-driven deployment procedures to optimize AWS infrastructure and enhance data analytical effectiveness.
- **Automated Certificate Lifecycle Management**: Developed and implemented automated certificate lifecycle management processes, including issuance, renewal, and revocation, to maintain a secure and compliant PKI infrastructure.
- Salesforce PKI integration:Integrated PKI with Salesforce Certificate and Key Management, to configure certificate-based authentication for Salesforce APIs, and to deploy certificates for secure communication.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer/Full Stack Developer(DF/Mulesoft/Java/Py)/Data Migration/Software Test Engineer
Responsibilities:

- Developed, maintained, and supported infrastructure solutions, enabling seamless data and workflow integration across AWS, Azure, and Salesforce Clouds (e.g., Sales, Service, Marketing) within multi-cloud environments.
- Developed and implemented ETL pipelines using AWS Glue, Python, and SQL to migrate Health Care Approved Providers' data into Salesforce, supporting a critical aged care sector transformation.
- Leveraged Amazon Redshift for high-performance data warehousing and analytics. Utilized Amazon EMR with PySpark to build data pipelines for extracting, transforming, and loading (ETL) data from diverse sources into a Data Lake. Implemented ETL processes using Apache Spark, integrating with AWS S3 for data storage and AWS Redshift for data warehousing.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- **Advanced Salesforce Support**: Delivered advanced Salesforce support, ensuring the seamless operation of critical university systems by proactively investigating and resolving complex technical challenges within OmniStudio components (FlexCards, OmniScripts, DataRaptors, Integration Procedures).
- **System Integrity and Administration**: Maintained robust system integrity through comprehensive administration, including user management, application metrics monitoring, configuration, and data integrity assurance."
- DevOps and Deployment Management: Implemented and managed deployments using a structured DevOps framework, contributing to efficient and reliable system updates.
- Troubleshooting and System Enhancement: Conducted thorough log analysis to diagnose and resolve system anomalies, enhancing system stability and performance. These efforts directly supported the university's ability to maintain efficient administrative & academic processes, ensuring reliable access to essential data & resources for faculty, staff, and students.
- Demonstrated ability to maintain and configure Salesforce Service Cloud and Health Cloud applications through declarative tools, ensuring data integrity and optimal system performance.
- Using either Python or Java, developed custom ETL/API orchestrations applicable to cloud and on-premise applications.
- Implemented automated testing frameworks (JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, AWS DevOps) to integrate development and testing processes. Utilized Git for version control, Maven and Gradle for building, & JUnit for automated testing.
- Implemented GitOps deployment strategy for efficient infrastructure and application management. Provisioned and managed AWS Kubernetes Service (EKS) clusters using Terraform and kubectl.
- PKI Integration and Testing: Designed and executed thorough testing of PKI integrations with cloud platforms and MuleSoft, validating certificate functionality and identifying potential security vulnerabilities.
- Incident Response and Monitoring: Developed and implemented incident response plans for PKI-related security incidents, and monitored certificate usage and potential security incidents.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Full Stack Developer(DF/Mulesoft/Java/Py)/Data Migration/Software Test Engineer
Responsibilities:

- Developed, maintained, and supported infrastructure solutions within multi-cloud environments, enabling seamless data and workflow integration between AWS, Azure, and specific Salesforce Clouds (e.g., Sales, Service, Marketing).
- Defined data requirements, structures, and transformation logic for complex data migration projects, resulting in streamlined data integration. Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Collaborated with stakeholders to gather & document specific business and data quality requirements for the target system.
- Developed a suite of mapping documents to define data transformations between multiple source and target systems.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.

- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis.
- **Docker Images:** Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Deployed Artifactory for secure artifact control & incorporated Jira for project oversight and issue resolution throughout the CI/CD lifecycle.
- **Automated Certificate Lifecycle Management**: Developed and implemented automated certificate lifecycle management processes, including issuance, renewal, and revocation, to maintain a secure and compliant PKI infrastructure.
- Salesforce PKI integration:Integrated PKI with Salesforce Certificate and Key Management, to configure certificate-based authentication for Salesforce APIs, and to deploy certificates for secure communication.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Software Test Engineer/ Data Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Developed, maintained, and supported infrastructure solutions within multi-cloud environments, enabling seamless data and workflow integration between AWS, Azure, and specific Salesforce Clouds (e.g., Sales, Service, Marketing).
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /EDW
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Designed, developed, and deployed efficient and scalable data integration solutions for on-premises data pipelines using Enterprise Ab Initio's ETL tools (GDE, ADC, Express-IT, Control Center) and advanced techniques (Meta Programming, Co>Relate). Implemented robust data quality features within Ab Initio-based data pipelines. Performed database backups (SQL, MongoDB, DynamoDB).
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.

- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, AI Agents, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.3, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Data Migration Engineer – NDIA via Acenture Aged Care Acnt-Paxus -10/3/2025 – Interviewed on 19/3/25
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Data Migration Engineer/Full Stack Developer /Integration Solution Architect /Software Engineer/ Python/R Snr DevOps Engineer Developer /Mulesoft /AI/ML/BI/SP/D 365 Developer /EDW/DBA –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Data Migration Engineer/Full Stack Developer(SF/MuleSoft/AI)/Snr DevOps/Salesforce Support Engineer
Responsibilities:

- Developed and implemented ETL pipelines using AWS Glue, Python, and SQL to migrate Aged Care Approved Providers' data into Salesforce, supporting a critical aged care sector transformation.
- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer:** Utilize Amazon EMR, a scalable managed Hadoop framework service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying Silver & potentially Gold layer of Data Lake for advanced analytics and reporting.
- Leveraged AWS Glue Studio for ETL, Apache Spark for data processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Leveraged AWS Glue Studio, Boto3, and Kinesis Data Streams to build a complex real-time data pipelines, ingesting data from a Python script-based generator, processing it using Glue Streaming ETL, and storing the results in Amazon S3.
- Defined data requirements, strFlex caructures, and transformation logic for complex data migration projects, resulting in streamlined data integration. Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Collaborated with stakeholders to gather & document specific business and data quality requirements for the target system.
- Developed a suite of mapping documents to define data transformations between multiple source and target systems.

- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Customized data writing logic to integrate with specific target destinations, such as Amazon Kinesis Data Streams, by utilizing GlueContext's write_dynamic_frame.from_options() method: glueContext.write_dynamic_frame.from_options()).
- Designed and implemented custom applications in Salesforce Sales Cloud, Service Cloud, and Health Cloud, utilizing declarative tools to automate and optimize aged care regulatory workflows. This included streamlining site audits, quality reviews, and assessment contacts. The applications actively generated Assessment Team Report & Performance Reports, which incorporated 42 Met/Not Met judgment KPIs (CSI, PCI, TFI), directly supporting accurate credit decision-making.
- Designed and implemented AI-driven CI/CD workflows using python, to automate the creation, prediction, and optimization of AWS IaC pipelines, using SageMaker, CloudFormation, and boto3: cfn = boto3.client('cloudformation') ...
- Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for data validation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- Automated repetitive tasks for Business Analysts and Data Analysts by developing secure, offline local AI systems (PrivateGPT-X, Ollama 3.3, Google NotebookLM, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- Modeled application data structures and storage solutions, contributing to robust and scalable data architectures.
- Developed and maintained CICD automation for ACQSC's Salesforce and MuleSoft, enabling faster releases and supporting the modernization of legacy systems via MuleSoft Anypoint APIs.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., dfx force:source:pull) from Salesforce organizations.
- **Automated Deployments with SFDX**: Minimized manual errors and accelerated releases. Enhanced Code Quality: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.Independently architecting a multi-region, fault-tolerant Kubernetes cluster to support critical applications.
- Provided expert-level Salesforce support by investigating and resolving complex issues within OmniStudio components (FlexCards, OmniScripts, DataRaptors, Integration Procedures), performing comprehensive system administration (user management, application metrics monitoring, configuration and data integrity), managing deployments through a DevOps framework, and conducting thorough log analysis to diagnose and resolve system anomalies.
- Developed and optimized SQL queries for data extraction, transformation, and loading (ETL) processes, emphasizing performance and data accuracy. Utilized the AWS SDK for Python (Boto3) to interact with Amazon Kinesis for real-time data streaming and processing. kinesis_client = boto3.client('kinesis', region_name='ap-southeast-2'
- Proficient in designing and implementing end-to-end automated processes using Salesforce Flow Builder, including record-triggered, screen, and scheduled flows, significantly enhancing operational efficiency.
- **Leveraged Kafka (MSK) Topics, MSK, producers, consumers, Kafka Connect, and Flink to build robust real-time data pipelines, enabling efficient ingestion, transformation, and processing of streaming data.**
- Tableau: Use Tableau to create interactive dashboards and visualizations to provide insights to business users.
- Amazon QuickSight: Leverage QuickSight for self-service BI and ad-hoc analysis.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Data Migration Engineer/Full Stack Developer(SF/MuleSoft/Java/Py/R)/Snr DevOps Engineer
Responsibilities:

- Engineered AWS Glue ETL pipelines with Python and SQL to migrate on-premises data into Salesforce, enabling a significant digital transformation within the RBA's financial services operations.
- **Architected and implemented a modern 3-layer lambda architecture on AWS:**
- **Batch layer: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes. Speed layer: Employed Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis.**
- Unified service layer**: Leveraged Amazon Kinesis Data Streams as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data warehouse for advanced analytics and reporting.**
- Leveraged Matillion and ERwin to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs, within both enterprise data warehouse (Inmon) and data mart (Kimball) architectures.
- Real-time Data Pipelines and Efficiency: Developed and implemented robust real-time data pipelines using AWS Glue Studio, Boto3, and Kinesis Data Streams, enabling the efficient ingestion and processing of streaming data from a Python-based generator, with secure storage in Amazon S3.
- Data Integrity and Accuracy: Defined data requirements, structures, and transformation logic for complex migration projects, ensuring streamlined data integration and upholding data integrity. Conducted thorough analysis of data sources, types, and formats to establish precise transformation, loading, and data model requirements, crucial for maintaining data accuracy."
- Stakeholder Collaboration and Documentation: Collaborated closely with stakeholders to gather and document comprehensive business and data quality requirements, ensuring alignment with organizational objectives. Developed detailed mapping documents to define intricate data transformations between diverse source and target systems.

- Implemented mapping logic with precision, utilizing both code and specialized tools to accurately map source data to target data according to defined rules, ensuring data reliability and consistency. This comprehensive approach to data management and migration supports the RBA's commitment to data-driven decision-making and operational excellence."
- Experienced in utilizing OmniStudio components (FlexCards, OmniScripts, DataRaptors, and Integration Procedures) to develop and maintain dynamic, user-friendly applications within the Salesforce ecosystem.
- Implemented and managed comprehensive CICD automation processes, significantly enhancing release efficiency for Aged Care's Salesforce and MuleSoft platforms, while simultaneously facilitating the transition of legacy systems to a modern, MuleSoft Anypoint API-driven architecture.
- Acted as a subject matter expert (SME) on data migration processes, providing guidance and support to team members.
- Customized data writing logic to integrate with specific target destinations, such as Amazon Kinesis Data Streams, by utilizing GlueContext's write_dynamic_frame.from_options() method. For instance, to write a dynamic frame to a Kinesis stream, the following Python code using AWS Glue and Boto3 can be implemented
- Automated Salesforce Development Workflow: Utilized Salesforce DX tools (dfx force:source:pull) to retrieve and manage code (metadata) from development orgs. Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using dfx force:source:deploy.
- Streamlined Salesforce Development and Deployment: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., dfx force:source:pull) from Salesforce orgs. Automated Deployments with SFDX: Minimized manual errors and accelerated releases. Enhanced Code Quality: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.Independently architecting a multi-region, fault-tolerant Kubernetes cluster to support critical applications.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Design, develop, and maintain automated ETL pipeline/API orchestration using either:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.
- Utilized Apache Spark SQL and DataFrame APIs to perform complex data transformations and aggregations. Leveraged AWS Glue Studio to create and maintain ETL jobs for data pipelines. Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch, along with Kafka and Spark for stream processing, applicable to cloud and on-premises applications.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Data Migration Engineer/Full Stack Developer(SF/MuleSoft/Java/Py/AI/R))/Snr DevOps Engineer
Responsibilities:

- Developed and implemented ETL pipelines using AWS Glue, Python, and SQL to migrate on-premises data into Salesforce, supporting a critical transformation of the University's enrollment and student services processes.
- Architected and implemented a modern 3-layer data warehouse architecture on AWS using Snowflake:
- Ingestion Layer (Replacing Batch Layer): Utilized Amazon S3 for staging raw data and Snowflake's Snowpipe for continuous, automated data ingestion. This layer focused on efficiently loading large volumes of data into Snowflake.
- Transformation Layer (Replacing Speed Layer): Employed Snowflake's powerful SQL engine and stored procedures for data transformation and cleansing, enabling real-time & batch processing. This layer focused on preparing data for analysis.
- Consumption Layer (Replacing Unified Service Layer): Leveraged Snowflake's virtual warehouses for optimized querying & data access, enabling advanced analytics & reporting through tools like Tableau and Power BI, directly querying Snowflake.
- Defined data requirements, structures, and transformation logic for complex data migration projects, resulting in streamlined data integration. Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between heterogeneous data systems and cloud environments.

- Utilized SPARX for data modeling to support efficient pipeline development within a comprehensive enterprise architecture framework. This data model can then be used for integration with Snowflake, AWS, SAS, and Python in either a Salesforce or Mulesoft development environment.
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift. glueContext = GlueContext(SparkContext.getOrCreate())
- Skilled in troubleshooting and optimizing OmniStudio components, including FlexCards, OmniScripts, DataRaptors, and Integration Procedures, to resolve complex business requirements.
- Implemented and managed robust validation rules, ensuring data accuracy and compliance across various Salesforce objects, contributing to improved data quality. validated_df = validate_data(cleaned_df.copy())
- Experienced with Salesforce deployments via DevOps framework, and monitoring application metrics.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Designed ETL processes using source-to-target mappings and design documents that incorporated security, performance tuning, and other best practices, including minimizing data input, implementing incremental data updates, maximizing data quality, and automating tasks.
- Implemented parallel processing, secured the data preparation area, and adhered to DevSecOps principles for secure development practices when building ETL data pipelines.
- Deploy & test code from lower environments to production. Document technical specifications, data models, process flows etc. Perform comprehensive ETL process and other technical components to ensure data quality and functionality.
- Optimize data delivery through automation and scalable infrastructure, evaluating process improvements.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis.
- **Docker Images:** Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Data Migration Engineer /DevOps Engineer/Software Test Engineer /Clouds/Linux Administrator/EDW
Responsibilities:

- Developed and implemented ETL pipelines using AWS Glue, Python, and SQL to migrate Health Care Approved Providers' data into Salesforce, supporting a critical aged care sector transformation.
- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting. kinesis_client = boto3.client('kinesis', region_name='ap-southeast-2')
- Leveraged Matillion and ERwin to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs, within both enterprise data warehouse (Inmon) and data mart (Kimball) architectures.
- Leveraged Amazon Redshift for high-performance data warehousing and analytics. Utilized Amazon EMR with PySpark to build data pipelines for extracting, transforming, and loading (ETL) data from diverse sources into a Data Lake. Implemented ETL processes using Apache Spark, integrating with AWS S3 for data storage and AWS Redshift for data warehousing.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- **Advanced Salesforce Support**: Delivered advanced Salesforce support, ensuring the seamless operation of critical university systems by proactively investigating and resolving complex technical challenges within OmniStudio components (FlexCards, OmniScripts, DataRaptors, Integration Procedures).
- **System Integrity and Administration**: Maintained robust system integrity through comprehensive administration, including user management, application metrics monitoring, configuration, and data integrity assurance."
- DevOps and Deployment Management: Implemented and managed deployments using a structured DevOps framework, contributing to efficient and reliable system updates.

- Troubleshooting and System Enhancement: Conducted thorough log analysis to diagnose and resolve system anomalies, enhancing system stability and performance. These efforts directly supported the university's ability to maintain efficient administrative & academic processes, ensuring reliable access to essential data & resources for faculty, staff, and students.
- Demonstrated ability to maintain and configure Salesforce Service Cloud and Health Cloud applications through declarative tools, ensuring data integrity and optimal system performance.
- Using either Python or Java, developed custom ETL/API orchestrations applicable to cloud and on-premise applications.
- Implemented automated testing frameworks (JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, AWS DevOps) to integrate development and testing processes. Utilized Git for version control, Maven and Gradle for building, & JUnit for automated testing.
- Implemented GitOps deployment strategy for efficient infrastructure and application management. Provisioned and managed AWS Kubernetes Service (EKS) clusters using Terraform and kubectl.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Data Engineer /DevOps Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer/Software Test Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Designed, developed, and deployed efficient and scalable data integration solutions for on-premises data pipelines using Enterprise Ab Initio's ETL tools (GDE, ADC, Express-IT, Control Center) and advanced techniques (Meta Programming, Co>Relate). Implemented robust data quality features within Ab Initio-based data pipelines. Performed database backups (SQL, MongoDB, DynamoDB).
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.

- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, AI Agents, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.3, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve

Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Principle DevOps Engineer-NDIA – Rubix -2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle DevOps Engineer/Snr Integration Developer/Snr Data Engineer/Java/.Net Developer/Clouds/Linux Administrator Salesforce /MuleSoft Developer /AI/ML/BI/SharePoint/D365 CRM Devloper/EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Principle DevOps Engineer-Salesforce & Azure/Snr Integration Developer/Py/R /AI/ML/SP/D365 CRM Developer
Responsibilities:

- **Leveraged MuleSoft Anypoint Platform for Hybrid Cloud Integration**: Designed APIs using MuleSoft's hybrid capabilities to seamlessly integrate between existing AWS infrastructure (API Gateway) and Azure IPaaS environments.
- Design & develop CI/CD pipelines for Salesforce deployments using Azure DevOps tools like Azure Pipelines, YAML &Bash
- **Automate Salesforce Environment Management**: Use IaC tools (Salesforce DX, Terraform, Ansible) to define configurations & automate provisioning, network setup, and deployments for Sandbox, Dev, and Production environments.
- **Automated Salesforce Development Workflow**:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
- Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx orce:source:deploy`.
- Design, deply, and maintain automated pipelines for Mulesoft deployments by Azure DevOps to deliver Mulesoft application.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.

- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **Streamlined Change Management**: Utilized Salesforce change sets (or a version control system like Git, GitLab, Azure Repos, or AWS CodeCommit) within the CICD flow for managing code changes.
- Leverage Git to manage Repos for version control and collaboration, while using Boards to manage the project workflow.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintet K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- **Integration Solution Design with Azure DevOps**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps Pipeline with source code management (e.g., Git, Azure Repos) to trigger builds for each integration platform (MuleSoft, Azure Logic Apps, BizTalk).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, Azure Resource Manager templates for Logic Apps/Azure Integration Services).
- **Streamlined Health Workflows**: Develop custom applications on the Salesforce platform (Sales Cloud, Service Cloud, etc.) tailored for the government regulator of aged care in health sector business (e.g. Plan & schedule Site Audit, Quality Review, Assessment contacts, CSI, PCI, TFI aged care business process workflows and Decision Makers used Assessment Team Report, Performance report with 42 Met, Not Met List judgment KPIs) for making Credit or Not Credit decisions.
- **Enhanced User Experiences with Vlocity and OmniStudio**: Build low-code/no-code user interfaces and streamline processes within the Salesforce CRM using Vlocity and OmniStudio. This allows for rapid development of custom portals, forms, and mobile experiences for NDIA participants/providers (NDIA) or patients/healthcare providers (Health).
- **Data-Driven Decision Making with Powerful AI in CRM Analytics**: Integrate CRM Analytics (formerly Tableau CRM) to transform Salesforce data into actionable insights using its most powerful AI features for leveraging capabilities like:
    - **Predictive Analytics**: Forecast future trends and potential issues related to participant needs, service utilization, or provider performance (NDIA) or patient demographics, treatment outcomes, resource allocation (Health).
    - **Einstein Discovery**: Automatically uncover hidden patterns and relationships within Salesforce data, leading to more informed recommendations and decisions for improved service delivery or resource management.
    - **Einstein Einstein Next Best Action**: Leverage AI to recommend the most effective next steps for NDIA staff (e.g., suggesting the most suitable plan for a participant) or healthcare providers (e.g., recommending the most appropriate treatment based on patient history).
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- **Deployed multi-cloud databases**: Utilized either Azure Cosmos DB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with Azure Cognitive Search, libraries like Faiss (available through NuGet packages), pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior DevOps Engineer/Snr Integration Developer/Snr Data Engineer/Java/AI/ML/ML/SP/BI Developer
Responsibilities:

- **Utilized MuleSoft Anypoint Platform for Hybrid Cloud Integration:** Designed APIs using MuleSoft's hybrid capabilities to seamlessly integrate between existing GCP infrastructure (Cloud API Gateway) and Azure IPaaS environments.
- Collaborate with developers, testers, and operations teams to ensure smooth deployments and efficient workflows.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the conectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Leverage Git to manage Repos for version control and collaboration, while using Boards to manage your project workflow.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintxt workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

- **Automated Salesforce Development Workflow**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for streamlined development and deployment. The pipeline utilizes `dfx force:source:pull` command to retrieve source code from a Salesforce org (e.g., `mydevorg`) using a specific username (`myuser@mydomain.com`).
- **Integration Solution Design with AWS CodePipeline**: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (MuleSoft, AWS Step Functions, or Hybrid).
- Implement robust version control practices using Git and conduct thorough code reviews of pull requests.
- **Develop and Build**: Use AWS CodeBuild project with source code management (e.g., Git, AWS CodeCommit) to trigger builds for each integration platform (MuleSoft, AWS Integration Services, BizTalk).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, AWS CloudFormation templates for Step Functions/Integration Services).
- **Automated Salesforce Development Workflow**:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
- Troubleshoot issues and maintain a high level of system availability
- Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx orce:source:deploy`.
- Design, deply, and maintain automated pipelines for Mulesoft deployments by Azure DevOps to deliver Mulesoft application.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.
- **Automated Metadata**: Streamlined metadata retrieval & deployment for efficient development and release cycles.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- Design, develop, and maintain a robust ETL pipeline/API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Performing data transformation and integration tasks either using Python in AWS CodeBuild to leverage the power of AWS CodePipeline to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Worked on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
- **Integrated DevSecOps practices with Infrastructure as Code (IaC) security principles** to automate security configurations and ensure consistent security posture across deployments.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Snr DevOps Engineer/Integration Developer/Snr Data Engineer/Python/ML/SharePoint/K2 Developer
Responsibilities:

- **Leveraged MuleSoft Anypoint Platform for Hybrid Cloud Integration**: Designed APIs using MuleSoft's hybrid capabilities to seamlessly integrate between existing AWS infrastructure (including API Gateway) and Microsoft Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminated the need for Terraform or complex infrastructure management, enabling greater agility and flexibility within the hybrid cloud environment
- Configure and manage Mulesoft environments (Development, Testing, Production) through infrastructure as code (IaC)
- **Automated Metadata**: Streamlined metadata retrieval & deployment for efficient development and release cycles.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- **Streamlined International Student Workflows**: Develop custom applications on the Salesforce platform (Sales Cloud, Service Cloud, etc.) tailored for managing international student needs at the University of Canberra. These applications could include functionalities like:
    - Online application portal for international students; Visa and immigration document tracking;
    - Accommodation and orientation program management; Course enrollment and academic progress monitoring
- These applications can leverage Salesforce DX for efficient development and deployment.
- Collaborate with developers, testers, and operations teams to ensure smooth deployments and efficient workflows.
- **Enhanced User Experiences with OmniStudio**: Build low-code/no-code user interfaces and streamline processes within the Salesforce CRM using OmniStudio. This allows for rapid development of custom portals, forms, and mobile experiences for international students at the University of Canberra. Examples include:
    - Self-service portal for international students to access resources, submit requests, and track applications;

    - Interactive mobile app for international students to access course information, campus maps, and connect with support services; Personalized communication channels to keep international students informed and engaged
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM 2013 using KingswaySoft SSIS Integration Toolkit for Microsoft Dynamics CRM 2013. Leverage Visual Studio Online, TFS, or Git for version control.
- Managing, supporting, and troubleshooting MuleSoft, Azure IPaaS, BizTalk, SharePoint 2019, Dynamics 365 CRM, and SQL Server 2012 on Windows Server 2012, with a strong focus on system and procedure documentation.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless failover between AWS and Azure. This ensures resilience against outages.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **Streamlined Change Management**: Utilized Salesforce change sets (or a version control system like Git, GitLab, Azure Repos, or AWS CodeCommit) within the CICD flow for managing code changes.
- Optimized code for ML projects using data structures, algorithms, and software architecture knowledge.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Analyzed text data using NLP techniques (text preprocessing, sentiment analysis, & topic modeling) with NLTK and Gensim.
- Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: DevOps Engineer /Integration Developer /Software Test Engineer/Cloud/Linux Administrator/DBA
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to your specific needs. Or
- **Java with Spring Batch along with Kafka and Spark** to provide powerful tools for managing complex data pipelines within your CI/CD process.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP, Bash & PowerShell scripting languages
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.

- Generated test plans, status reports, and summaries to keep stakeholders informed.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Systems Integration Developer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Design & develop CI/CD pipelines for Mulesoft deployments using Azure DevOps tools like Azure Pipelines, YAML & Bash
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored Azure cloud resources with Virtual Machines and Blob storage using tools like Azure Monitor, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized Azure Monitor for logging, partitioning, indexing, & optimizing Azure Data Factory structure to analyze system & application performance, providing insights for business case development to enhance service delivery & reduce issues.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in Azure using Azure Functions, Logic Apps, Azure API Management, Azure Service Bus Queues, Azure Event Grid, Azure Monitor, Resource Manager, and Azure Blob Storage, resulting in improved system efficiency.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Managed Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Implement robust version control practices using Git and conduct thorough code reviews of pull requests.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Salesforce Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.

- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010~Nintex K2 Five Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer –EKS-eBet Gaming System – No agent -2/3/2025 – Failed notice by Seek, not eBet on 5/3/25
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Integration Solution Architect /Software Engineer/Snr Data Engineer /Linux/Clouds Administrator/ Java/.NET/Salesforce/MuleSoft/Python AI/ML/DL/BI/SharePoint/D365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Snr Clouds/Systems Administrator/Python/AI/ML/SharePoint/D365 Developer /EDW
Responsibilities:

- Designed and implemented a secure hybrid cloud architecture integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted site-to-site connections using Cloud VPN Gateways across all platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Architected and deployed secure, scalable AWS Landing Zones for diverse workloads, utilizing IaC (CloudFormation/ Terraform) and AWS Control Tower for centralized governance.

- Automated infrastructure provisioning and configuration of network resources via Python (Boto3), Ruby with Chef, or Ansible, enhancing efficiency and consistency. Implemented efficient and scalable container orchestration with ECS Fargate and cross-account observability through Amazon CloudWatch, ensuring system reliability and performance.
- Linux Server Provisioning: Employed Ruby with Chef cookbooks (e.g., ec2_instance.rb, network_config.rb, package_install.rb, deploy_app.rb) for automated EC2 instance creation, network configuration, package installation, and application deployment.
- Leveraged Ruby within Chef cookbooks to automate Nginx web server deployments and AWS EC2 instance configurations, integrated with Git version control and GitHub Actions workflows for seamless CI/CD pipelines within AWS environments.
- Utilized Embedded Ruby (ERB) templates within Chef cookbooks to dynamically generate and deploy Nginx configuration files (e.g., '/etc/nginx/sites-available/default'), automating Nginx service restarts upon template modifications.
- Implemented Embedded Ruby (ERB) templates in Chef cookbooks to dynamically generate & deploy Nginx configuration files (e.g., '/etc/nginx/sites-available/default'), ensuring automatic Nginx service restarts upon template modifications.
- Ensured robust security and compliance of AWS environments through the effective use of CloudTrail and AWS Config.
- Using Chef to define infrastructure (servers, software, configurations, etc.) as code, security features, and Policyfile workflow to automate infrastructure configuration & ensure consistent, secure, & scalable deployments.
- Document processes and best practices to boost operational efficiency, aligning with the ITIL 4 framework and Agile development methodologies. **Collaborate with development and operations teams to streamline deployments.**
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Chef for configuration management and deployments on artifact images.
- Continuously researched and implemented innovative DevOps solutions, focusing on enhancing CI/CD pipeline efficiency and infrastructure automation using cutting-edge techniques and relevant industry trends.
- Designed and implemented containerized application migrations to AWS ECS Fargate, utilizing Docker, optimizing performance and cost efficiency.
- Built and managed AWS ECS clusters and task definitions to support automated CI/CD pipelines, ensuring seamless application deployment and management.
- Implemented robust CI/CD pipelines utilizing GitHub Actions for automated builds, testing, and deployment to ECS Fargate.
- Implemented highly scalable and fault-tolerant monitoring solutions, utilizing Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to track metrics, logs, and events from applications and infrastructure, including advanced log analysis and troubleshooting with CloudWatch Insights.
- Implemented custom CloudWatch metrics using Boto3 to track critical application performance indicators, enabling data-driven decision-making and proactive issue resolution: cloudwatch = boto3.client('cloudwatch')
- Python or Go programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Designed and maintained automated CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps, Azure Pipelines**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy, Cloud Composer**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Automated performance monitoring and vulnerability scanning within CI/CD pipelines using Datadog and Nessus, enabling rapid identification and resolution of performance bottlenecks and security threats. Implemented security best practices based on ISO 27001 and ACSC Essential 8, contributing to a robust and compliant system environment
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Design and implement a proactive fraud detection system, using risk control AI agents to identify & mitigate potential threats
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT-X, Ollama3.3, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Snr DevOps Engineer/Snr Clouds/Systems Administrator /Java/Python /Go/AI/ML/BI/SP Developer /EDW
Responsibilities:

- Developed a secure hybrid cloud architecture, establishing encrypted connections between AWS VPCs, GCP VPCs, and Azure VNets via Cloud VPN Gateways. Implemented infrastructure-as-code using Terraform or Ruby with Chef for automated, consistent, and efficient provisioning and configuration.
- Streamlined infrastructure management by utilizing Python (Boto3), Ruby with Chef, or Ansible for automated provisioning and configuration of network resources, resulting in improved efficiency and consistency. Implemented robust container orchestration using AWS ECS and established cross-account monitoring through Amazon CloudWatch, enhancing system reliability and performance.

- Automated Nginx web server deployments and AWS EC2 instance configurations using Ruby within Chef cookbooks, integrating with Git for version control[8][9]. Implemented GitHub Actions workflows to create seamless CI/CD pipelines within AWS environments, optimizing the development and deployment process
- Developed dynamic Nginx configuration file generation using Ruby ERB templates within Chef cookbooks[9][10]. This approach enabled automatic updates to files such as '/etc/nginx/sites-available/default' and ensured Nginx service restarts upon template modifications, improving system flexibility and responsiveness
- Cloud & DevOps: Deploy ML pipelines on AWS ECS, SQS, S3, Lambda, Step Functions) with CI/CD automation using GitHub Actions.
- Designed, implemented, and optimized CI/CD pipelines using GitHub Actions workflows defined in main.yml (or build.yml). Automate infrastructure with Chef –a pull configuration management tool to ensure seamless system performance. Troubleshoot and resolve build, deployment, and system configuration issues.
- Linux Server Provisioning: Automated EC2 instance creation, network configuration, software installation, and application deployment using Ruby with Chef cookbooks and recipes (e.g., ec2_instance.rb, network_config.rb, etc.).
- **Automation and CI/CD**: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision): model = torch.hub.load('pytorch/vision:v0.6.0', 'mobilenet_v2', pretrained=True)
- Expertise in implementing and maintaining a robust suite of cloud security controls, including Identity and Access Management (IAM), data encryption, network security, vulnerability management, and data loss prevention (DLP). Ensured compliance with industry standards and regulations such as CIS, NIST, GDPR, and HIPAA.
- Designed, developed, and maintained automated ETL/ELT pipeline/API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban).
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision: model = models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
- Applied technical expertise in monitoring, observability, performance management, risk management, troubleshooting, and service recoverability. Executed and reported on non-functional testing, contributing to system improvements.
- Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
- Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.

Oct. 2013 – Nov.2015 University of Canberra (UC)**, ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer/Snr Clouds/Systems Administrator/Java/.NET/Python/Mulesoft/SP/K2 Developer
Responsibilities:

- Cloud Architecture: Designed and implemented a hybrid cloud architecture seamlessly integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets, facilitating interoperability and resource optimization across platforms.
- Configured encrypted tunnels for data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

- Automated network infrastructure provisioning and configuration with Python, Ruby, and Ansible, significantly improving efficiency. Deployed and managed scalable containerized applications on AWS ECS, ensuring system reliability through CloudWatch observability.
- Built automated Nginx and EC2 deployment workflows using Ruby (Chef) and integrated them into AWS CI/CD pipelines with Git and GitHub Actions.
- Linux Server Provisioning: Leveraged Ruby with Chef recipes (e.g., ec2_instance, network_setup, install_tools, deploy_service) to automate EC2 instance provisioning, network configuration, software installation, and application deployments.
- Automated Kubernetes deployments using Helm, streamlining the management of containerized applications built with Docker.
- Utilized Embedded Ruby (ERB) templates within Chef cookbooks to dynamically generate and deploy Nginx configuration files (e.g., '/etc/nginx/sites-available/default'), automating Nginx service restarts upon template modifications.
- Designed and maintained automated CI/CD pipelines using platform-specific tools:
    1. **Azure: Azure DevOps, Azure Pipelines**
    2. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    3. **GCP: Cloud Build/Cloud Deploy, Cloud Composer**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Proven ability to design & implement highly scalable, resilient, & cost-effective cloud-native applications on the Azure platform, leveraging technologies such as Azure Service Bus, Azure Event Hubs, Azure Container Service (ACS), and Azure DevOps.
- **Secured API Access for Student Enrollment**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.
- **CI/CD and DevSecOps Automation:** Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- Utilized Terraform/CloudFormation (or similar) to automate the provisioning of cloud infrastructure for a PySpark-based data platform, ensuring consistent and reproducible deployments.
- Integrated AWS ECS service into the data platform CI/CD pipeline, automating containerized deployments and ensuring seamless integration with AWS infrastructure.
- Automated ETL/API orchestration using Python (Airflow) or Java (Spring Batch), implementing CI/CD pipelines for seamless deployment, monitoring, and infrastructure management.
- Automated AWS IaC pipeline creation and optimization using Python, SageMaker, CloudFormation, and boto3, emphasizing CI/CD best practices and infrastructure as code principles.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps Engineer/Snr Clouds/Systems Administrator /Software Test Engineer/Linux Administrator
Responsibilities:

- Deployed and secured a hybrid cloud architecture, connecting AWS, GCP, and Azure networks with encrypted VPN tunnels. Automated infrastructure provisioning & configuration using Terraform or Ruby with Chef, ensuring consistency & efficiency.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- Developed and maintained automation scripts using Python, Go, Bash to streamline development & operational workflows.

- **Leverage AWS services for cloud-native design patterns**:
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI/CD, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Successfully leveraged ITIL 4.0 frameworks, including Incident Management, Problem Management, and Continual Improvement, to optimize incident response processes and drive service excellence.
- Linux Server Provisioning: Utilize Ansible playbooks (e.g., ec2_instance.yml, configure_router.yml, network_config.yml, install_packages.yml, deploy_app.yml) for creating EC2 instances, configuring networks, installing packages, and deploying applications.
- Automated Windows Server administration using Chef, configuring Active Directory, group policies, and software deployments. Utilized knife bootstrap windows_server_ip -r 'recipe[your_recipe]' to initialize Chef clients on Windows servers, ensuring consistent configurations.
- Database Provisioning: Automated database creation and user management using Puppet. Developed Puppet manifests to create databases (e.g., mysql::db { 'mydatabase': ensure => 'present', }) and manage database user permissions.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Snr Clouds/Systems Administrator /Software Test Engineer/SP Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Streamlined infrastructure management by utilizing Python (Boto3), Ruby with Chef for automated provisioning and configuration of network resources, resulting in improved efficiency and consistency. Implemented robust container orchestration using AWS ECS and established cross-account monitoring through Amazon CloudWatch, enhancing system reliability and performance.
- Automated Nginx web server deployments and AWS EC2 instance configurations using Ruby within Chef cookbooks, integrating with Git for version control. Implemented GitLab CI/CD workflows to create seamless CI/CD pipelines within AWS environments, optimizing the development and deployment process
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- Linux Server Provisioning: Automated EC2 instance creation, network configuration, software installation, and application deployment using Ruby with Chef cookbooks and recipes (e.g., ec2_instance.rb, network_config.rb, etc.).

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH

Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, AI Agents, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.3, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

F.it Talenza Martin Senior Systems Engineer (Azure/InTune/Defender) – Mirvac – Talent Martin -21/2/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter

Senior Systems Engineer (Azure/InTune/Defender)/Senior Clouds/Linux Administrator/Senior DevOps /Data Engineer / Java/.NET/Python/AI/ML/BI/SharePoint/Dynamics 365 Consultant/Develper /EDW –Japan (7.5yrs) & Australia (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Systems Engineer(Azure/InTune/Defender)/Snr DevOps/Data Engineer /Python/AI/ML/SP/D365 Developer
Responsibilities:

- Architected and implemented a hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented secure Azure landing zones for Linux and Windows workloads, leveraging Azure Blueprints and incorporating security best practices. Integrated these landing zones with M365 services, including MS Teams, SharePoint Online, Dynamics 365, Power BI, Power Automate, and Power Apps, ensuring secure access and data governance.
- **Azure IaaS**: Successfully deployed, managed, and maintained Azure IaaS solutions, including compute (Virtual Machines, Scale Sets, AKS), storage (Blob, Queue, Disk), networking (VNet, Load Balancer, VPN Gateway), and security services (Azure AD, Security Center, Key Vault) to support critical business applications.
- **Azure Cloud Optimization**: "Optimized Azure cloud environments for cost efficiency, performance, and security using tools like Azure Cost Management + Billing, Azure Advisor, Azure Monitor, Azure Automation, and Azure Security Center, resulting in 20% cost reduction, 30% performance improvement.
- Collaborated with stakeholders to gather requirements and design network solutions that met business needs, including secure connectivity between multiple sites and Azure cloud environments.
- Designed, implemented, and managed site-to-site VPN connections between on-premises Meraki MX security appliances and Azure Virtual Networks, ensuring secure and reliable communication between corporate networks and cloud resources.
- Configured and maintained Meraki MX security appliances, including firewall rules, VPN settings, and other network configurations, to optimize network performance and security. This involved configuring & managing security features such as intrusion detection/prevention (IDS/IPS), content filtering, & malware protection, effectively acting as a network defender.
- Connect Intune and Defender for Endpoint together and create policies to scan file, detect threats, and report threat levels. Configure CA policies to block access from non-compliant devices.
- Developed and implemented fully automated CI/CD pipelines using Azure DevOps, Azure Pipelines or GitHub Actions.
- **Cloud Infrastructure Engineering: Azure Expertise**: Proficient in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, firewall rules and intrusion prevention systems, SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure Entra ID (formerly known as AAD)-based access control for Azure Key Vault.
- **Network Engineering**: Led design and deployment of robust network solutions, leveraging deep knowledge of CIDR, TCP/IP stack, OSPF, RIP, and BGP for optimal performance & security across diverse WAN & hybrid cloud environments.
- **SD-WAN** (Software-Defined Wide Area Network): Implemented an SD-WAN solution using Cisco Meraki, VeloCloud across 10 branch offices, resulting in a 30% reduction in WAN costs and improved application performance.
- Managed and configured a range of Azure services, including Virtual Machines, Virtual Networks, Storage Accounts, Azure SQL Database, and Azure Active Directory, to support critical business applications.
- Successfully migrated device management tasks from SCCM to Microsoft Intune using Hybrid Azure AD Connect for user synchronization and Hybrid Azure AD Join for device management.
- Troubleshot and resolved network connectivity issues, including VPN connectivity problems, to minimize downtime and ensure business continuity. Managed and refreshed network infrastructure, including firewalls, Cisco switches, and WAPs(Wireless Access Points), following best practices and SDLC processes
- Maintained up-to-date documentation of network configurations and VPN settings to facilitate knowledge sharing and efficient troubleshooting. Prioritized security controls and implemented measures to protect network infrastructure and data, including securing VPN connections and enforcing firewall policies.
- **Patch management**: Use Windows Autopatch with Intune to automate Windows 10/11 updates, and SCCM for server patches. Leverage PowerShell scripting for additional automation and customization as needed.
- **Device configuration**: Use Intune for mobile device management and SCCM for on-premises device configuration.
- **Autopilot Deployment Profile**: Create a deployment profile that specifies the desired configuration settings for the devices. Using Intune to enroll devices into Azure AD either using QR codes, bulk enrollment, or Autopilot deployment profiles.
- Managed and administered both Windows and Linux-based systems across multiple locations, demonstrating a versatile skillset in infrastructure management. Provided support and maintenance for Linux servers hosting web applications, databases, and other critical services, ensuring high availability and performance.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Design and implement a proactive fraud detection system, using risk control AI agents to identify & mitigate potential threats
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT-X, Ollama3.2, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- Provided day-to-day ICT technical support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Using InTune and Configuration Service Providers(CSP) for mobile device management (MDM) on Windows devices.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Snr Systems Engineer(Azure/InTune/Defender)/Snr DevOps/Data Engineer AI/ML/SP/BI Developer/EDW
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Built and implemented Medallion Architecture with Data Lake in Azure**:
- Batch Layer: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- Serving Layer: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- Real-time Layer: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- **Azure IaaS**:Engineered and maintained robust and secure Azure IaaS solutions, encompassing compute (Virtual Machines, Scale Sets, AKS), storage (Blob, Queue, Disk), networking (VNet, Load Balancer, VPN Gateway), and identity and access management (AAD, Security Center, Key Vault) to ensure the high availability and integrity of critical business applications.
- Optimized Azure cloud environments for cost-effectiveness and performance, leveraging tools like Azure Cost Management + Billing, Azure Advisor, Azure Monitor, and Azure Automation. Achieved a 20% reduction in operational expenditure and a 30% improvement in application performance while maintaining stringent security standards.
- Configured and maintained Meraki MX security appliances, implementing robust firewall rules, VPN settings, and network configurations to ensure network security and compliance with regulatory requirements. This included deploying and managing advanced security features such as intrusion detection/prevention (IDS/IPS), content filtering, and malware protection to proactively defend against cyber threats.
- Integrated MS Intune and Defender for Endpoint to enhance endpoint security and enforce compliance with organizational policies. Developed and implemented policies for file scanning, threat detection, and threat level reporting. Configured Conditional Access policies to restrict access from non-compliant devices, ensuring a secure & controlled endpoint env.
- Engineered and maintained secure Azure cloud infrastructure, demonstrating expertise in designing, implementing, and managing Virtual Networks (VNets), subnets, Network Security Groups (NSGs), firewall rules, Intrusion Prevention Systems (IPS), SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure Entra ID (formerly Azure AD)-based access control for Azure Key Vault. Prioritized security best practices and compliance requirements throughout the infrastructure lifecycle.
- Led the design and deployment of highly resilient and secure network solutions, leveraging deep expertise in CIDR, TCP/IP stack, OSPF, RIP, and BGP. Optimized network performance and security across diverse WAN and hybrid cloud environments, implementing redundancy and failover mechanisms to ensure business continuity.
- SD-WAN: Implemented and managed an SD-WAN solution utilizing Cisco Meraki and VeloCloud across 10 branch offices, resulting in a 25% reduction in WAN costs and demonstrable improvements in application performance. This initiative enhanced network agility and simplified branch office connectivity while maintaining robust security standards.
- Administered and maintained a range of secure Azure services, including VMs, Virtual Networks, Storage Accounts, Azure SQL Database, and AAD, ensuring the high availability and integrity of critical business applications. Implemented security best practices and access controls to protect sensitive data and maintain compliance with regulatory requirements.
- Modernized device management by successfully migrating from SCCM to Microsoft Intune. Leveraged Hybrid Azure AD Connect for user synchronization and Hybrid Azure AD Join for device management, enhancing security and streamlining IT operations. This transition improved device compliance and enabled more efficient management of endpoint devices.
- Network Connectivity & Troubleshooting: Ensured high network availability and business continuity by proactively troubleshooting and resolving network connectivity issues, including VPN disruptions. Minimized downtime through rapid response & effective problem-solving, using established incident management procedures.
- Managed and refreshed network infrastructure, including firewalls, Cisco switches, and Wireless Access Points (WAPs), adhering to industry best practices and established Software Development Lifecycle (SDLC) processes. Prioritized security hardening and compliance throughout the infrastructure lifecycle.
- Automated patch management for Windows 10 clients and servers, leveraging Windows Autopatch with Intune and SCCM, respectively. Utilized PowerShell scripting to enhance automation and customize patching processes, ensuring timely updates and minimizing security vulnerabilities.
- Utilized Intune for mobile device management and SCCM for on-premises device configuration.

- Streamlined device deployment and enhanced security by implementing Autopilot profiles and leveraging Intune for automated Azure AD enrollment. Utilized QR codes, bulk enrollment, and Autopilot deployment profiles to ensure consistent device configurations and enforce security policies from the point of deployment. This approach minimized manual intervention and reduced the risk of misconfigurations.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Systems Engineer(Azure/InTune/Defender)/Snr DevOps/Data Engineer /ML/SPoint /K2 Developer /DBA
Responsibilities:

- Cloud Architecture: Designed and implemented a hybrid cloud architecture seamlessly integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets, facilitating interoperability and resource optimization across platforms.Secure Data Transfer:
- Configured encrypted tunnels for data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented secure Azure landing zones for Linux and Windows workloads, leveraging Azure Blueprints and incorporating security best practices. Integrated these landing zones with M365 services (MS Teams, SharePoint Online, Dynamics 365, Power BI, Power Automate, Power Apps), ensuring secure access and robust data governance.
- **Azure IaaS Management**: Deployed, managed, and maintained Azure IaaS solutions, encompassing compute (Virtual Machines, Scale Sets, AKS), storage (Blob, Queue, Disk), networking (VNet, Load Balancer, VPN Gateway), and security services (Azure AD, Security Center, Key Vault) to support mission-critical applications.
- **Azure Cloud Optimization**: Optimized Azure cloud environments for cost efficiency, performance, and security, utilizing tools like Azure Cost Management + Billing, Azure Advisor, Azure Monitor, Azure Automation, and Azure Security Center. Achieved a 20% cost reduction and a 30% performance improvement.
- Network Solution Design: Collaborated with stakeholders to gather requirements and design network solutions that met business needs, including secure connectivity between multiple sites and Azure cloud environments.
- **Secure Site-to-Site VPN**: Designed, implemented, and managed secure site-to-site VPN connections between on-premises Meraki MX security appliances and Azure Virtual Networks, ensuring secure & reliable communication between corporate networks and cloud resources.
- **Meraki MX Security Management**: Configured and maintained Meraki MX security appliances, implementing firewall rules, VPN settings, and other network configurations to optimize network performance and security. Deployed and managed security features like intrusion detection/prevention (IDS/IPS), content filtering, and malware protection.
- **Endpoint Security Management**: Integrated Microsoft Intune and Defender for Endpoint, creating and implementing policies for file scanning, threat detection, and threat level reporting. Configured Conditional Access policies to block access from non-compliant devices, ensuring a secure endpoint environment.
- Automated CI/CD Pipelines: Developed and implemented fully automated CI/CD pipelines using Azure DevOps, Azure Pipelines, and/or GitHub Actions, accelerating software delivery and improving release quality.
- **Azure Infrastructure Engineering**: Designed, implemented, and managed Azure infrastructure, including VNets, subnets, NSGs, firewall rules, intrusion prevention systems, SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure Entra ID-based access control for Azure Key Vault.
- **Network Engineering Expertise**: Led the design and deployment of robust network solutions, leveraging expertise in CIDR, TCP/IP stack, OSPF, RIP, and BGP for optimal performance and security across diverse WAN & hybrid cloud environments.
- **SD-WAN Implementation**: Implemented an SD-WAN solution using Cisco Meraki and VeloCloud across 5 branch offices, achieving a 20% reduction in WAN costs and improved application performance.Azure Service Administration: Managed and configured a range of Azure services, including Virtual Machines, Virtual Networks, Storage Accounts, Azure SQL Database, and Azure Active Directory, to support critical business applications.
- Device Management Migration: Successfully migrated device management tasks from SCCM to Microsoft Intune using Hybrid Azure AD Connect for user synchronization and Hybrid Azure AD Join for device management.Network
- **Troubleshooting and Maintenance**: Troubleshot and resolved network connectivity issues, including VPN connectivity problems, minimizing downtime and ensuring business continuity. Managed and refreshed network infrastructure, including firewalls, Cisco switches, and WAPs, following best practices and SDLC processes.
- Maintained up-to-date documentation of network configurations & VPN settings. Prioritized security controls & implemented measures to protect network infrastructure and data, including securing VPN connections and enforcing firewall policies.
- Automated Patching: Automated patch management for Windows 10 clients and servers using Windows Autopatch with Intune and SCCM for server patches. Leveraged PowerShell scripting for additional automation and customization.
- **Device Configuration Management**: Managed device configuration using Intune for mobile device management and SCCM for on-premises device configuration.
- **Autopilot Deployment**: Created and implemented Autopilot deployment profiles for standardized device onboarding. Used Intune to enroll devices into Azure AD using QR codes, bulk enrollment, or Autopilot deployment profiles.
- Managed and administered both Windows and Linux-based systems across multiple locations, providing support and maintenance for Linux servers hosting web applications, databases, and other critical services.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS, and PCI-DSS frameworks.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**